Progressive Retrieval Attention for Pretrained Transformers: Sparse Native-K/V Memory with Semantic Routing

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Abstract

Dense long-context attention couples logical memory size to the key/value (K/V) state active in every transformer operation. PRA-HF separates these quantities: it searches a compact semantic index and materializes only selected layer-native K/V under a fixed attention budget. At the measured pinned Qwen, SmolLM2 Llama-family, and official Gemma 3 1B checkpoints, the verified PRA-off paths leave logits, hidden states, greedy outputs, and the tested cache contracts exactly unchanged.

Controlled positional analysis in companion Paper 1.5 motivates the empirical rule *route semantic state and attend native positional state*. On frozen Qwen3-0.6B, a 262,144-parameter router reaches any-evidence recall 0.831 at 10% and 0.956 at 20% selected parents while returning exact post-RoPE K and native V. Oracle HotpotQA replay is depth dependent: the last 14 layers gain 3.641 nats per gold token, whereas all-layer replay loses 0.766. An end-to-end audit finds complete evidence coverage and exact cached K/V, ruling out a hidden oracle-path bug. Oracle replay nevertheless remains distinct from native context: co-materialized non-evidence tokens receive .310 of normalized late-band Hotpot attention support, source-only K/V differ from question-contextualized K/V, and harmful all-layer replay coincides with substantially greater residual divergence than late-band replay. PRA-conditional adapters preserve the ordinary no-memory path exactly and leave the measured WikiText-2 loss unchanged. On QASPER, decoded polarity improves from 25.0% with routed frozen PRA to 62.5% with oracle evidence and $72.5 \pm 5.6\%$ with residual 16, separating evidence selection from memory consumption; blind behavioral evaluation independently favors residual adaptation. Simple decision-token reweighting does not generalize in this small-data regime. HotpotQA provides a harder multi-hop boundary: frozen routed PRA remains positive at 6.56% active K/V, but dominant relation-near misses and routed-adapter brittleness motivate iterative retrieval. The official Gemma path preserves its native local/global organization and shows a positive routed likelihood effect. The shared execution core, thin family adapters, router artifacts, wheel, and Python API make this cross-family boundary directly testable.

1 Introduction

Dense long-context attention couples logical memory size to the K/V participating in each transformer operation. Retaining more history therefore increases both stored state and active attention, even when most of that history is irrelevant to the current token. Prompt truncation breaks the coupling by discarding history. Text retrieval narrows the prompt before inference, but then re-encodes the selected text. PRA instead keeps already computed model state outside the active context and restores only selected native K/V.

Paper 1 established bounded sparse native-K/V memory in controlled models [32]. Paper 1.5 separated source coordinates, contextual fidelity, native attention geometry, and semantic routing geometry [31]. This paper asks whether those boundaries can be retrofitted into pretrained Hugging Face decoders without changing their native attention semantics. That requirement is strict. Qwen, Llama-family, and Gemma decoders differ in projection details, RoPE helpers, grouped-query layout, masks, cache protocols, and output paths. An adapter can produce plausible text while violating one of these contracts.

Retrofitting also exposes two distinct interfaces. A routing function R_ϕ answers *what memory should be active?*; a memory-use function M_ψ answers *how should that memory affect frozen computation?* The state best suited to finding a semantically relevant chunk need not be the state a pretrained attention head expects after selection. PRA-HF therefore separates a compact *routing index* from the layer-native *detail K/V* payload, then confines any learned memory-use correction to the PRA-active branch. Paper 1.5 supplies the controlled scientific motivation; this paper tests transfer and implementation in pretrained models. The resulting rule is: **route semantic state; attend native positional state.**

The paper series separates mechanisms that otherwise become easy to conflate. Paper 1 separates logical memory from active K/V; Paper 1.5 separates positional and attention geometry from semantic routing geometry; this Paper 2 tests exact pretrained retrofit, causal use, and family portability. Paper 2.5 will isolate whether the multi-hop boundary arises from entry discovery, associative propagation, or bounded competition while keeping the Paper-2 memory-consumption path fixed. Paper 3 will study progressive or partial K/V materialization after routing.

Figure 1 gives the complete data path before the individual contracts are developed. Publication creates a small search representation and a full native payload; only the selected payload enters active attention.

FROZEN PRETRAINED MODEL WITH EXTERNAL NATIVE-K/V MEMORY

large logical source → model-safe encoding blocks → native post-RoPE K / native V

↘ compact hidden-state gists → R_ϕ: "what memory?"

current hidden state ↗ selected chunk identities

↓ exact whole-chunk materialization

↓ bounded native attention

↓ memory-induced update d_l

M_ψ: "how should memory affect computation?"

↙ ↘

delta-only residual adapter | ⊕ if jointly selected | PRA-conditional output LoRA

↓ frozen downstream transformer path → logits

Figure 1: Canonical PRA-HF path and its two learned interfaces. **PRA-HF compresses the search representation, not the selected memory payload.** R_ϕ selects memory from compact semantic state. Native attention consumes exact selected positional K/V. M_ψ is optional, PRA-conditional, and structurally absent when external memory is disabled.

Table 1: What Paper 2 establishes. Each answer is tied to a distinct evaluation stage; success in one row does not imply success in the next.

Question	Answer
Can PRA retrofit a pretrained HF decoder without altering its disabled path?	Yes: logits, hidden states, greedy output, and tested cache contracts have exact parity.
Can compact semantic state select useful native K/V?	Yes, especially on QASPER; the learned router reaches any-evidence recall .831/.956 at 10/20%.
Does perfect annotated-evidence discovery reproduce native context?	No: exact oracle selection and exact cached K/V still leave a functional oracle gap.
What remains between discovery and behavior?	Contextual representation, materialized support, depth-sensitive integration, decision formation, and decoding.
Can light adaptation help?	Yes on QASPER: residual 16 improves likelihood, polarity, containment, EOS, and blind preference, but not universally.
Does the mechanism transfer across families?	Execution, routing, and bounded functional probes are demonstrated on Qwen, SmolLM2, and Gemma with stated limits.
What is the SDK default?	Frozen PRA plus the validated learned router; residual 16 is QASPER-specific and rank-32 LoRA remains research-only.

The paper makes five contributions. First, it defines and tests an exact pretrained-model retrofit for bounded native-K/V memory, including exact disabled-path parity. Second, it separates compact semantic routing state from the selected layer-native positional K/V payload. Third, it measures sparse discovery as a recall–sparsity frontier and shows that a tiny supervised projection improves frozen semantic routing. Fourth, it establishes depth-dependent causal memory use, audits why oracle replay differs from native context, and separates a clean-memory integration ceiling from the routed end-to-end robustness boundary. Fifth, it implements the contract as a shared core, thin Qwen, Llama-family, and Gemma adapters, router artifacts, accounting, a wheel, and a public Python API. The contribution is a working boundary, not a new pretrained model or a solved long-context QA system.

The datasets deliberately have different jobs (Table 2); averaging them would erase the distinction between preservation, one-shot document use, and multi-hop evidence gathering.

Table 2: Dataset roles in Paper 2. Each regime has a different success criterion.

Dataset	Primary role	What success means
WikiText-2	No-memory preservation	PRA-conditional specialization leaves ordinary loss unchanged
QASPER	One-shot long-document memory	Sparse routed memory improves likelihood and judged output utility
HotpotQA	Multi-hop stress and oracle causal probe	Exposes the boundary that motivates iterative evidence gathering

2 PRA-HF in 60 Seconds

A long source is encoded in blocks that fit the host model. Publication keeps compact semantic gists in a routing index and stores exact layer-native post-RoPE keys and native values as the detail payload, which may remain on CPU. A query searches only the compact index. Selected chunk identities pass through a hard budget, and only their native K/V moves to the accelerator for bounded attention with the direct sequence.

This division answers the most important implementation question. PRA does *not* attend to the gist. The gist is an address label, analogous to a compact catalog entry. Once an address is chosen, attention reads the original native K/V payload. Consequently, routing can use a semantic hidden state without asking a pretrained attention head to consume an unfamiliar representation. Likewise, positional correction is a routing concern only when positional keys are used as gists; the selected attention payload always retains its original native rotary encoding.

The budget distinguishes logical availability from physical activation. A reference may contain thousands of cached tokens, while one operation materializes only B selected tokens plus the direct sequence and a small safety reserve. The mechanism therefore bounds active attention even when the logical source is longer. It does not make storage free: full layer-native detail K/V still occupies CPU memory unless a separate compression or tiering method is added.

Table 3: Eight evaluation stages for memory-augmented prediction. They are diagnostic stages, not a guaranteed causal sequence; success at one does not imply success at the next.

Stage	Question	Primary evidence	Measured status
Availability	Was source state published and addressable?	Publication, lifecycle, and accounting	Named external memory outside active attention
Discovery	Were useful memory identities found sparsely?	Recall–sparsity	Strong in-domain; weak cross-domain
Representation	What contextual state did selected sources acquire?	Aligned direct/source K/V	Measurably different from native direct context
Materialization	Which native K/V became active, and was transport exact?	Parity, identity, K/V, and budget audits	Exact when disabled and for selected payload; bounded when active
Integration	Did memory causally alter frozen computation?	$\Delta \log p$; depth; residuals	Positive in a depth-dependent band
Decision	Did memory support the task-critical branch?	Polarity; margin; rank	Improved on QASPER after conditional adaptation
Decoding	Was the answer realized and terminated correctly?	F1; EM; containment; EOS	Improved but still incomplete under routed memory
Behavioral utility	Was output semantically equivalent or preferable?	Blind LLM judges	Positive QASPER signal; evaluator-dependent HotpotQA

Adaptation is not a ninth computational stage. It is an intervention that can change integration, decision, or decoding while the evaluation stages remain fixed.

3 Minimal PRA Recap

For layer ℓ, PRA stores reference chunks as native $K_c^{(\ell)}, V_c^{(\ell)} \in \mathbb{R}^{H_{kv} \times T_c \times d_h}$ and one or more compact gists $g_c^{(\ell)}$. A query representation $r^{(\ell)}$ scores the gists, and a global budgeter selects whole chunks.

Materialized memory precedes the direct K/V sequence under one softmax:

$$\mathrm{Attn}(Q, [K_M; K_D], [V_M; V_D]) = \mathrm{softmax}\left(\frac{Q[K_M; K_D]^\top}{\sqrt{d_h}} + M\right)[V_M; V_D]. \tag{1}$$

This is not a separately normalized cross-attention residual. The native output projection is applied once to the combined result.

Paper 1 established controlled native-K/V routing and materialization [32]. Paper 1.5 established that source position, contextual fragmentation, pooled native-attention geometry, semantic relevance, and downstream use are distinct [31]. Continuous sources use $p_i = p_{\text{logical start}} + i$, and ordinary post-RoPE keys remain the canonical cache state. The present adapter preserves that contract by capturing each selected HF layer's actual post-RoPE keys and by assigning source-relative positions across bounded encoding blocks.

4 Exact Retrofitting: Shared Core, Thin Adapters

4.1 Two representations, two jobs

PRA represents a cached chunk as

$$\mathcal{M}_c = (R_c, K_c, V_c), \tag{2}$$

where R_c is a compact routing representation and (K_c, V_c) is the payload used after selection. The objects need not inhabit the same feature space. The canonical HF configuration in this paper sets R_c to one or more gists of the hidden state entering the selected attention block, while K_c is that block's native post-RoPE key tensor and V_c its native value tensor. Token-level hidden states are transient during publication; only compact gists and detail K/V persist.

This split follows the specialization chain

$$h_{\text{in}} \longrightarrow W_K h_{\text{in}} \longrightarrow \mathrm{RoPE}(W_K h_{\text{in}}). \tag{3}$$

The first representation retains broad contextual information, the second is an attention-address representation, and the third conditions that address on token position. Native token attention needs the third. Chunk retrieval may prefer the first. The routing query therefore uses the final attention-input hidden state and is compared only with hidden-state gists. It is never compared directly with Q_{post}. This is an empirical default for the tested Qwen checkpoint, not a claim that hidden states are universally optimal routers.

4.2 Shared execution core

The shared core divides the former monolithic attention path into explicit stages. Query preparation selects the newest valid state and maps GQA query groups into native K/V routing width. Exact packed-index search returns parent identities. The materialization budget then enforces

$$L_{\text{direct}} + L_{\text{materialized}} + L_{\text{reserve}} \le L_{\text{native,max}}. \tag{4}$$

`materialize_selected_kv` removes duplicate overlap, preserves row isolation, and moves only retained detail K/V. `combine_local_and_memory_kv` pads variable rows and prepends an additive all-visible memory mask to the family's existing local causal mask. The controlled model and HF adapters now call this same core.

4.3 Thin HF contract

The abstract adapter exposes six family operations: project Q/K/V, apply native position encoding, normalize tensor layout, combine the native mask, invoke PRA attention, and project the output. Disabled memory takes an even narrower path: call the original attention object unchanged. This design preserves pretrained parameters and avoids checkpoint conversion.

The Qwen adapter has 272 physical lines (248 nonblank, non-comment lines). It introduces no learned parameter. Its family-specific branches distinguish Qwen2-style projections from Qwen3's Q/K normalization; routing, budgeting, materialization, residency, and metrics remain outside the file. This is a portability measurement, not a code-minimization target.

The Gemma adapter is narrower. It reuses the same projection and capture path but rejects any layer whose native `is_sliding` flag is true. The wrapper republishes this metadata because `Gemma3DecoderLayer` chooses local versus global rotary embeddings before calling its attention child. Local layers therefore retain their 512-token window, local RoPE base, mask, and hybrid-cache behavior. At global layers, the adapter invokes Gemma's installed eager kernel with its attention-logit soft cap and leaves the native one-head MQA payload unexpanded in storage. No Gemma branch was added to routing, budgeting, materialization, or reference publication.

This organization suggests a compatible hybrid pattern: native local computation plus PRA-enabled global integration. More cautiously, the host model's learned attention-scope structure provides a natural prior for external-memory placement. This is consistent with the Qwen result that sparse external memory should not be injected indiscriminately, but it does not establish a universal placement rule or show that the measured Gemma placement is optimal.

The exact target is `google/gemma-3-1b-it` at revision `dcc83ea841ab6100d6b47a070329e1ba4cf78752`. Its documented contract is 26 layers, four query heads, one K/V head, width 256 per head, and a five-local/one-global periodic schedule; global layers are 5, 11, 17, and 23 under zero-based indexing. Tests instantiate Transformers' real Gemma 3 classes with the same hybrid semantics and require bit-exact disabled logits, hidden states, greedy tokens, and every cache tensor. The official checkpoint passes those gates. Reference payloads remain one-head native MQA on CPU. Architecturally, layers 5, 11, 17, and 23 are global-attention capable; the routed experiment configures layers 17 and 23, those same two layers actually consume selected memory, and layer 23 provides routing state. A 384-token logical prompt is processed through 128-token bounded operations with zero native-limit violations.

4.4 Qwen native semantics

4.4.1 Projection and RoPE

The adapter calls the wrapped module's `q_proj`, `k_proj`, and `v_proj`. Qwen3's existing `q_norm` and `k_norm` are applied before the installed Transformers `apply_rotary_pos_emb` function. Reference capture saves post-RoPE K and position-independent V. No independent RoPE implementation is introduced.

4.4.2 Grouped-query attention

The tested checkpoint has $H_q = 16$, $H_{kv} = 8$, and $d_h = 128$. Cache objects retain $[B, 8, T, 128]$ K/V. Key-space routing baselines average each pair of query heads sharing one K/V head, producing a vector in the same 8×128 space as a flattened key gist. The canonical hidden-state router instead uses one d_{model} vector and does not alter payload head layout. K/V expansion occurs inside Qwen's eager attention function, never in persistent PRA storage. A synthetic replay test compares this path with explicit K/V head repetition and matches within 10^{-6}.

4.4.3 HF cache and masks

Prefill and single-token decode both use the wrapped module's HF `Cache.update` call. PRA memory is prepended after that update, so direct generation history appears once. A decode test with four prompt tokens and two generated tokens observes five direct cache positions at the last invocation, plus four separate memory positions. The original causal mask remains on local/cache positions; selected historical memory receives a visible prefix mask. Memory- disabled generation delegates entirely to the original module.

5 Evaluation Design

Table 4 records the first checkpoint exactly. Tests use synthetic Qwen3 configs offline. The pre-trained run uses Python 3.10.11, PyTorch 2.12.1+CUDA 12.6, Transformers 4.55.4, eager attention, FP16, and an NVIDIA GeForce GTX 950M. The frozen model is not fine-tuned.

Table 4: Pinned Qwen checkpoint and PRA placement.

Checkpoint	`Qwen/Qwen3-0.6B`
Revision	`c1899de289a04d12100db370d81485cdf75e47ca`
Parameters / layers	596,049,920 / 28
Query heads / K/V heads / head width	16 / 8 / 128
RoPE base / advertised positions	10^6 / 40,960
Initial PRA placement	layer 27 only
Depth-study instrumentation / selected band	all 28 layers / layers 20–27
Initial gist / routing policy	mean, one gist/chunk, exact top-k
Native limit / direct / selected-memory cap	256 / 128 / 64 tokens
K/V residency	CPU full cache, selected transfer

Table 5: Recall metrics answer different discovery questions and are never substituted for one another.

Metric	Meaning
Any-evidence recall	Fraction of examples for which at least one annotated evidence region is selected
Evidence-identity recall	Fraction of all annotated evidence identities recovered
All-evidence recall	Fraction of examples for which every required annotated evidence identity is selected

Every run writes JSON and scalar CSV with the Git SHA, model revision, software versions, device, dtype, layer selection, limits, routing settings, timings, and metrics. The exact executed commit is retained in each artifact rather than inferred from the manuscript build.

6 Exactness and Bounded Execution

6.1 Disabled-path parity

PRA-disabled parity is the hard gate. The baseline is evaluated first; the same in-memory model is then wrapped, leaving every parameter object in place. Table 6 reports exact results for the five-token prompt "The capital of Portugal is" and four-token greedy decode.

Table 6: Original HF model versus memory-disabled PRA wrapper.

Metric	Result
Maximum / mean absolute logit difference	0 / 0
Maximum hidden-state difference	0
Greedy token agreement	100%
Greedy sequences exactly equal	yes
Generation-cache shapes equal	all 28 layers
Finite adapted logits	yes

Single-run prefill times were 0.447 s before wrapping and 0.136 s after wrapping. This order is confounded by first-call warm-up and is not a speed comparison. Exact outputs, rather than latency, are the parity evidence.

6.2 Native-K/V reference transport

An explicit 22-token reference about Lisbon is encoded through the frozen model. The selected layer captures $[1, 8, 22, 128]$ K/V, stores it on CPU, creates a mean gist, and transfers all 22 tokens because the source fits one routing chunk. The physical transfer is 90,112 bytes, exactly $2 \times 22 \times 8 \times 128 \times 2$ bytes for FP16 K and V. The run reports 1.6 ms routing, 0.40 ms materialization orchestration, 0.15 ms selected transfer, and 0.11 ms combined attention. These synchronized phase values are instrumentation checks on one machine, not throughput claims. Cold reference construction took 0.144 s and the warm query took 0.191 s.

The generated continuation contains "Lisbon", but the unmodified model also knows this fact. Consequently, this observation establishes functional selected-K/V transport and generation, not a causal retrieval benefit. The exact native-replay claim is instead restricted to the synthetic tensor test, where GQA memory plus local K/V equals explicit dense composition.

6.3 Implicit head and bounded long prompt

A repeated prompt tokenizes to 408 logical tokens. The direct cap retains the final 128 tokens; the first 280 become `#_head`. That head is encoded in blocks of at most 64 tokens, with source-relative positions. The direct tail receives positions 280 through 407. Every encoding operation and the final selected-memory operation stays under the 256-token configured native limit; the largest observed source-encoding call is 64 tokens and violations are zero. The final finite-logit query took 0.258 s. This proves bounded execution and offset propagation for the frozen Qwen adapter. It does not show that the router selected the information needed by an arbitrary continuation.

6.4 Transport does not establish discovery

The transport tests intentionally do not establish semantic selection. In one fixed HotpotQA example [37] and one QASPER example [9], question-only, dense-truncated, oracle text-RAG, and PRA conditions diagnose plumbing rather than accuracy. PRA routes three late-source chunks in both cases and selects no annotated evidence; Qwen also fails the oracle text-RAG controls. The functioning K/V path therefore coexists with an unsuitable post-RoPE mean-key routing representation. This negative smoke test motivates the semantic routing/native-payload separation evaluated next; full records remain in the released artifacts.

7 Semantic Routing, Native Attention

Controlled experiments in Paper 1.5 show that increasingly accurate approximations of native Q/K geometry do not necessarily improve semantic evidence retrieval: centered $G = 1 \to 8$ raises Q/K rank fidelity from .794 to .963 while semantic AUC remains .6510–.6516, whereas a learned 32-D relevance geometry reaches .971 AUC with only .178 Q/K fidelity [31]. We therefore treat the routing representation and native attention payload as separate interfaces and ask whether this separation transfers to an unmodified pretrained Qwen model.

Controlled principle from Paper 1.5

native positional-attention geometry $\neq$ semantic routing geometry

$\Downarrow$ pretrained transfer

attention-input hidden query/gist $\longrightarrow$ tiny learned relevance projection

$\downarrow$ selected parent identities

$\downarrow$ exact layer-native post-RoPE K / native V

$\downarrow$ pretrained native attention

Figure 2: **Route semantic state; attend native positional state.** Paper 1.5 motivates the interface separation; Paper 2 tests it in pretrained models. Routing returns identities, while the attention path materializes the host layer's exact native K/V.

The code makes this independence executable. Routing returns stable *selected chunk identities*; a separate core entry point applies the common budget and materializes their payload. A fixed-selection test supplies the same identity set with ordinary-router and oracle metadata. It obtains bit-identical post-RoPE K, identical V, and exactly identical native attention output. Router choice can change the set, but it cannot silently change what a fixed set means to attention.

7.1 Pretrained transfer: what should represent a chunk?

The transfer test compares three parameter-free summaries on 16 matched examples, eight from HotpotQA and eight from QASPER. All use 32-token parents and exact top-k search. Post-RoPE key means retain explicit positional structure; pre-RoPE key means remove that phase from the index; attention-input hidden means retain broader contextual state. The learned row uses the separate 32-identity held-out cohort and five router seeds described in the next section. It is included to show the resulting design progression, not as a paired fourth row. In every condition, a fixed parent identity materializes the same post-RoPE native K and native V.

Table 7: Qwen routing-representation transfer. The first three rows use 16 matched diagnostic examples; the learned row uses 32 held-out identities and reports the five-seed 95% interval at R@3. Score-position r is correlation with normalized source position, and recall is any-evidence recall.

Routing state	Position r	R@3	R@8	R@16	Interpretation
Post-RoPE key mean	.652	.125	.250	.313	Strong positional structure; weak semantic index
Pre-RoPE key mean	.009	.313	.563	.750	Explicit phase removed; incomplete relevance geometry
Hidden-state mean	-.021	.438	.500	.750	Best zero-shot low-budget recall; contextual state
Learned asymmetric-128	-.026	$.631 \pm .069$	.806	.963	Task-specific relevance geometry

The controlled Paper 1.5 geometry separation transfers to a pretrained transformer. Pooled post-RoPE K retains strong positional structure and is a poor semantic routing index in Qwen. Removing explicit RoPE phase nearly eliminates measured position bias but does not itself define the best relevance geometry. Attention-input hidden state is independent of the subsequent K rotation and gives the strongest zero-shot low-budget result. This progression motivates the 128-D router rather than another positional repair.

A centered-RoPE Qwen control reproduces the same dissociation between native-Q/K fidelity and semantic retrieval. Its G sweep, implementation details, parameter-free recall curves, and segment, token, and query ablations are reported in Appendix G. The main result is the transferred interface boundary, not a new RoPE derivation.

8 Sparse Discovery with a Tiny Learned Router

Parameter-free cosine similarity asks the pretrained hidden space to serve a new retrieval objective without supervision. We instead freeze all 596,049,920 Qwen parameters and learn only two projections,

$$s(q, c) = \mathrm{norm}(W_q h_q)^\top \mathrm{norm}(W_c g_c), \tag{5}$$

where h_q is the last query state and g_c is one mean hidden-state gist. The selected asymmetric 1024-to-128 linear router contains 262,144 parameters, 0.044% of Qwen. Training uses 48 examples, validation uses 16, and the held-out test uses 32, with no QA-identity overlap. Validation selects an exhaustive pairwise margin objective; five optimization replicates over the same held-out identities measure the selected configuration.

On the held-out cohort, frozen last-state cosine reaches any-evidence R@3/R@8/MRR of 0.469/0.719/0.413; the learned router reaches 0.631 ± 0.069/0.806/0.498. HotpotQA and QASPER R@3 are 0.588 and 0.675. The paired R@3 gain is 0.163 ± 0.069, position bias is negligible ($r = -0.026$), and the matched shuffled-label control collapses to 0.125 R@3. Supervision, rather than low dimension alone, exposes the improved relevance geometry. The router is not yet general: Hotpot-to-QASPER and QASPER-to-Hotpot transfer both reach only 0.213 any-evidence recall@3. Wider projections, sampled negatives, and one hard-negative refresh do not improve the validation-selected result; Appendix G reports these controls.

8.1 Recall–sparsity, not fixed rank, defines the operating point

Any-evidence recall@3 is an intentionally severe diagnostic, but it changes meaning as source length changes. For each example with N_i parents, we also select $k_i(f) = \lceil f N_i \rceil$. Historical artifacts support *any-evidence recall*: whether at least one annotated parent is recovered. Complete rankings support the stricter *evidence-identity recall*: the fraction of all mapped evidence parents recovered. We never substitute one definition for the other.

Table 8: Any-evidence recall versus requested parent fraction. The learned row averages five seeds and 160 seed–example evaluations. Other rows are descriptive historical cohorts.

Mechanism	Any R@5%	Any R@10%	Any R@20%	Any R@30%	AUC_{0--30}
Post-RoPE key mean	0.125	0.125	0.250	0.375	0.190
Pre-RoPE key mean	0.375	0.563	0.938	1.000	0.700
Hidden-state mean	0.438	0.625	0.625	0.813	0.574
Last-state query	0.531	0.719	0.875	0.906	0.724
Learned margin router	**0.675**	**0.831**	**0.956**	**1.000**	**0.835**

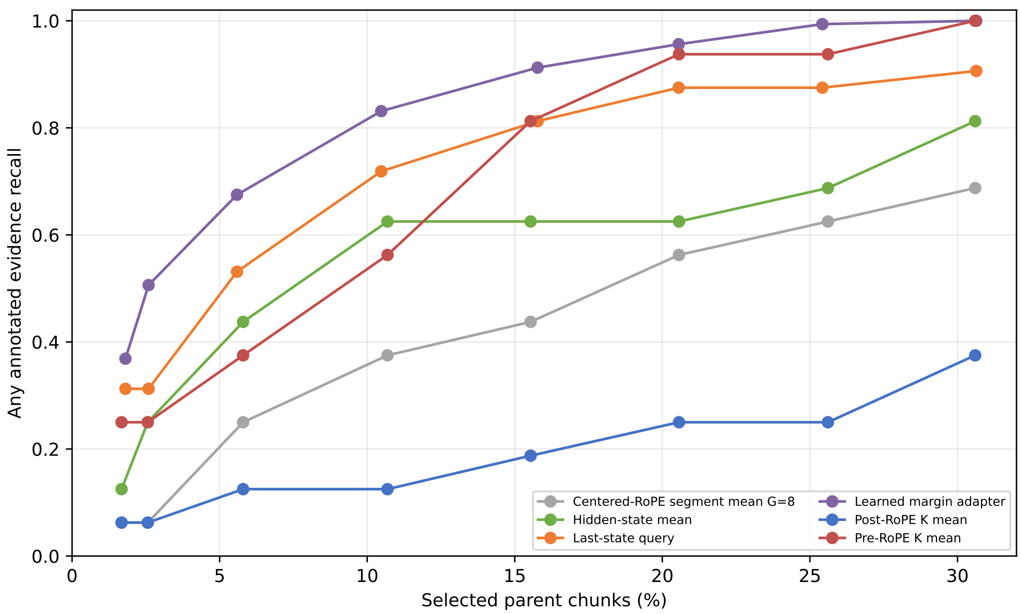

Figure 3: **The tiny learned router moves the recall–sparsity frontier.** Any-evidence recall rises sharply in the 10–20% selected-parent range that fixed-rank evaluation conceals. Cohorts differ for historical baselines, so only the learned result supports uncertainty over seeds.

At requested 10% and 20%, the selected route-once seed recovers 0.356 and 0.442 of annotated evidence identities, while its corresponding any-evidence values are 0.844 and 0.906. These are not the five-seed any-evidence means of 0.831 and 0.956 in Table 8. The seed, cohort, and metric differ. Thus a system can often find one useful passage while still missing other evidence required for a multi-hop answer. This distinction explains why HotpotQA needs a larger realized fraction for 90% any-evidence recall (0.207) than QASPER (0.104).

Scaling must also hold the denominator fixed. In the standalone predecessor, selecting a fixed eight references makes the selected fraction shrink from about 12.5% to 3.1% as memory grows from 64 to 256 units; apparent coverage consequently falls. Selecting approximately 12.5% instead keeps HotpotQA-derived target coverage at 0.756/0.772/0.762 and QASPER-derived coverage at 0.794/0.803/0.800 for 64/128/256 units. Ranking remains stable at a fixed fraction, while physical materialization grows with that fraction. Sparse quality and active-K/V cost must therefore be reported together.

9 From Availability to Behavioral Utility

Routing recall is necessary, but it is only one functional gate. We organize the evidence with the same eight-stage hierarchy used throughout the paper:

$$\boxed{\begin{array}{r}\text{Availability} \to \text{Discovery} \to \text{Representation} \to \text{Materialization} \\ \to \text{Integration} \to \text{Decision} \to \text{Decoding} \to \text{Behavioral utility}\end{array}} \tag{6}$$

Availability asks whether published source state can be accessed without placing the whole source in active attention. Discovery asks whether the right parent identities are found. Representation asks what K/V those parents acquire when encoded outside the live question. Materialization asks which native K/V tokens from the selected parents become active under the hard budget. Transport fidelity is tested at that boundary: it asks whether the selected cached payload survives exactly, not whether source-only K/V equal the state formed in native direct context. Integration measures the resulting change inside the frozen transformer. Decision tests whether that change supports the task-critical output branch. Decoding tests lexical realization and termination; behavioral utility tests semantic equivalence or preference. Thus

$$\boxed{\text{transport fidelity} \neq \text{representation fidelity} \neq \text{functional utility.}} \tag{7}$$

Training an adapter is an intervention on one or more stages, especially integration, decision, and decoding; it is not an additional stage.

9.1 Depth-dependent integration

The first causal question is whether correctly selected native K/V affects the model, and where. We replace learned selection by annotated-evidence oracle selection while keeping publication, materialization, attention, prompts, and the frozen backbone unchanged. Table 9 reports the comparison a reader needs from the full placement sweep in Appendix A.

Table 9: Oracle memory is depth sensitive. Values are mean gold-sequence $\Delta \log p$ versus the no-memory condition on four fixed examples per dataset. Direct evidence is ordinary prompt text, not PRA memory.

Dataset	Oracle, last 14	Oracle, all 28	Direct evidence text
HotpotQA	+3.641	-.766	+4.635
QASPER	+.072	-7.492	-1.791

Hotpot establishes a late-layer compatibility band: evidence K/V is causally useful, yet exposing the same selected memory in every layer reverses the benefit. QASPER's direct-text control is negative on this small slice, so it cannot establish annotation sufficiency there; its strongest oracle placement is the last eight layers at +.347. These are mechanistic probes rather than benchmark estimates.

9.2 Perfect discovery does not imply native-context equivalence

The all-layer regression prompted an end-to-end audit, not another retrieval mechanism. The historical cohort reproduces with maximum absolute difference zero. Every representable annotation is covered, requested and retained identities agree, all 28 Qwen layers are exercised, native-limit violations are zero, and freshly recomputed versus cached post-RoPE K and native V have maximum absolute error zero. Oracle discovery and payload transport are therefore correct, but oracle replay still differs functionally from placing evidence in the prompt.

Table 10: Oracle-gap explanations after the audit. "Supported" identifies a measured contributor, not a causal variance decomposition.

Candidate explanation	Paper-2 evidence	Status
Oracle implementation bug	Exact historical reproduction; 100% representable-evidence coverage; 28/28 layers; zero cached-K/V error	Rejected
Contextual K/V mismatch	Late-band pre-RoPE K cosine .940/.963 and V cosine .892/.926 for Hotpot/QASPER; encoding-length effects differ by dataset	Supported, dataset dependent
Softmax/support dilution	Hotpot late-band attention mass .220 evidence, .310 co-materialized non-evidence, .469 local/head state	Supported on Hotpot
Annotation insufficient for the model	Hotpot exact evidence +4.635 and parent context +5.574; QASPER exact and parent controls both -1.791	Rejected on Hotpot; unresolved on this QASPER slice
Missing early exposure	All 28 layers receive verified oracle K/V, yet all-layer replay is worse	Insufficient explanation

The .310 result has a precise scope: under oracle evidence selection, co-materialized non-evidence tokens receive approximately 31.0% of the normalized late-band attention support partitioned among evidence tokens, those non-evidence tokens, and ordinary local/head state on the four-example Hotpot diagnostic cohort. Per-example means have median .328; three of four exceed .25 and one exceeds .50. Those descriptive values show dispersion, but four examples do not support a population estimate. We call these tokens *co-materialized non-evidence tokens*, not semantic distractors. The result establishes

$$\text{oracle selection} \neq \text{evidence-only attention support.} \tag{8}$$

Removing that support only in an offline softmax counterfactual raises evidence mass from .220 to .281; it is not a partial-materialization experiment. Native direct evidence nevertheless succeeds with only .120 evidence mass, below either PRA condition. Attention mass is therefore neither a necessary nor a sufficient scalar explanation of functional utility.

Representation and integration remain separate as well. The source-only publication path does not reproduce the question-conditioned hidden trajectory used by direct context. Increasing encoding length modestly improves the one-example Hotpot probe but moves QASPER in the opposite direction despite higher cosine similarity. As in Paper 1.5's geometry study, representational closeness is not itself a sufficient proxy for functional utility.

Finally, harmful all-layer replay coincides with substantially greater final residual-stream divergence than late-band replay: relative L2 rises from .541 to .869 on Hotpot and from .602 to 1.245 on QASPER. This association does not establish that divergence causes the likelihood regression. It does show that "more layers receive exact memory" is not equivalent to "memory becomes more native." Figure 4 places the functional, representation, support, and divergence measurements together; the full span, layer, head, K/V, and encoding receipts remain in Appendix A.

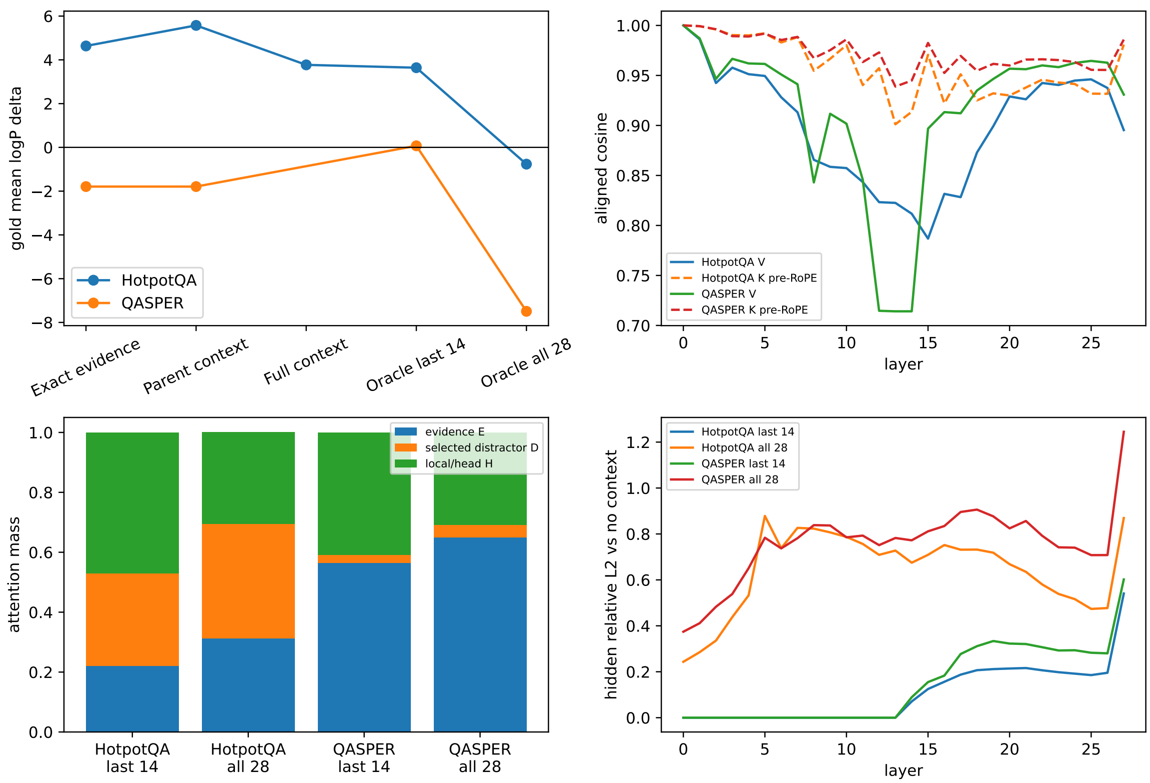

Figure 4: The oracle gap remains after exact discovery and transport. Direct-context controls, aligned K/V similarity, attention support, and residual divergence diagnose distinct stages; no panel alone is presented as a complete causal explanation.

Paper 2 holds the materialized parent payload fixed and diagnoses measurable attention allocated to co-materialized non-evidence state. Paper 3 intervenes on that variable directly through evidence-centered and adaptive native-K/V materialization under matched selection.

9.3 QASPER is the positive one-shot regime

The oracle audit should not obscure the paper's strongest pretrained result. QASPER supports one-shot semantic discovery, positive routed sequence likelihood, and improved decoded behavior under a small PRA-conditional residual adapter. Table 11 keeps the primary decision and completion outcomes together. The likelihood and blinded-behavior protocols use the same frozen test identities but different fixed generation instruments; they are convergent measurements, not one pooled score.

Table 11: QASPER's primary positive result. Polarity, containment, and EOS use the 32-token diagnostic. Sequence gain and blind preference come from their separately frozen protocols. Residual-16 values aggregate five seeds.

Condition	Routed $\Delta \log p$	Polarity	Containment	EOS	Blind preference
Frozen routed PRA	+1.377	.250	.500	.875	2.5%
Frozen oracle PRA	–	.625	.625	.625	–
Residual 16	+4.942	$.725 \pm .056$	.825	.950	22.5%

Residual 16 also improves routed F1 from .073 to .205 in the frozen behavioral package, and the two blind judges agree strongly by QASPER dimension ($r = .991$ for equivalence and $r = .995$ for relative quality). Exact PRA-off parity and unchanged measured WikiText-2 loss show that this memory-specific specialization does not alter the tested no-memory path. A larger rank-32 LoRA raises routed QASPER sequence benefit to +4.967, but its routed Hotpot change is -8.640 nats; accordingly, it remains research-only. Residual 16 is the strongest balanced QASPER condition, while frozen PRA plus the router remains the generic SDK default. Full adaptation, completion, decision-loss, and blinded-judge receipts appear in Appendices A, H, and I.

HotpotQA plays a different role. Its outputs are commonly relation-near but fail to reach the required endpoint, and one-shot semantic routing does not reliably complete multi-hop evidence chains. Paper 2.5 isolates whether this boundary arises from entry discovery, associative propagation, or bounded competition while keeping the Paper-2 memory-consumption path fixed.

9.4 Portability and practical boundary

The same shared execution contract runs through thin Qwen, SmolLM2 Llama-family, and official Gemma 3 adapters. All three measured checkpoints preserve the disabled path exactly and use the same 128-dimensional router interface; five-seed QASPER identity recall at 20% is $.454 \pm .007$, $.447 \pm .051$, and $.432 \pm .025$, respectively. Gemma preserves its native sliding/global organization, consumes PRA memory only in global layers 17 and 23, and obtains a positive routed likelihood effect of +.135 with unchanged F1. This demonstrates execution, routing-interface, and bounded functional portability, not equal quality or universal placement. Detailed family and systems tables are in Appendices B and C.

The released `pra-hf` package provides reference publication/removal, fraction-based selection, serializable router artifacts, bounded accounting, a wheel, CLI, notebook, and public Python API. In the measured product path, approximately 20% requested parents become 6–7% physical active K/V after whole-parent rounding and hard caps. These are correctness and reusability results from an eager implementation, not serving-speed claims. The public contract and code-level reimplementation path are detailed in Appendix J.

10 Limitations and Series Handoffs

The evidence establishes mechanisms on modest cohorts, not benchmark accuracy. Router seeds measure optimization sensitivity, not independent test examples. QASPER is a one-shot long-document regime, Hotpot is a multi-hop stress test, and WikiText-2 checks no-memory preservation. The oracle-depth and oracle-gap cohorts contain four examples per dataset; their attention and divergence measurements are diagnostic. Blind judges improve semantic coverage but are not ground truth. No experiment supplies a fused-kernel or matched dense-serving baseline. Extended protocol limits and all negative controls are retained in Appendix D.

The paper series assigns the remaining interventions to distinct variables. Paper 2.5 isolates whether the Hotpot boundary arises from entry discovery, associative propagation, or bounded competition while keeping the Paper-2 memory-consumption path fixed. Paper 3 changes materialization through evidence-centered and adaptive native-K/V payloads. Paper 4 studies systematic static internal stability. Paper 2 neither anticipates their results nor mixes those interventions into the pretrained-retrofit claim.

11 Related Work in Context

PRA-HF sits between text retrieval, sparse/long-context attention, KV retention and offload, latent memory, and parameter-efficient adaptation. Unlike text RAG, it reuses selected model-native state rather than re-encoding selected text; unlike ordinary sparse attention, it keeps named memory objects outside the active sequence; unlike KV compression, it compresses the search representation while retaining selected payload exactly; and unlike recurrent memory, it does not repeatedly transform anonymous state. RAG, RETRO, Longformer, Reformer, Memorizing Transformers, landmark attention, LoRA, and related systems define the neighboring design space [23, 6, 3, 21, 35, 27, 17]. The full comparison and series ownership map appear in Appendix E.

12 Conclusion

Paper 2's central result is a decomposition rather than a single score:

$$\begin{aligned} \text{availability} &\neq \text{discovery} \neq \text{representation} \neq \text{materialization} \\ &\neq \text{integration} \neq \text{decision} \neq \text{decoding} \neq \text{behavioral utility}. \end{aligned} \tag{9}$$

PRA-HF makes source state available outside the prompt, discovers parents through compact semantic gists, represents them through independently encoded native K/V, materializes the exact selected payload under a hard budget, and preserves the disabled host path exactly across three decoder families. QASPER shows that one-shot routed memory and light conditional adaptation can improve likelihood and decoded behavior while preserving ordinary language. Hotpot establishes the multi-hop discovery boundary.

Perfect discovery still does not reproduce native context. The oracle audit verifies exact selection and exact cached K/V, then identifies contextual-state mismatch, substantial Hotpot attention on co-materialized non-evidence tokens, and a larger internal departure under harmful all-layer replay. These findings keep representation, materialization, and integration as separate computational problems. Frozen PRA plus the validated router is therefore the conservative SDK default. Paper 2.5 next isolates entry discovery, associative propagation, and bounded competition; Paper 3 separately intervenes on what selected memory is materialized.

Appendix guide. The appendices preserve, in order, oracle/depth/adaptation details; cross-family receipts; product and systems measurements; extended limitations; detailed related work; the prior full scientific synthesis; routing ablations; QASPER decision-loss controls; the behavioral-judge protocol; and the code-level reimplementation guide. Main-text claims cross-reference the corresponding receipt rather than repeating its full diagnostic ladder.

A Oracle, Depth, Adaptation, and Behavioral Details

Retrieval recall asks whether annotated evidence was selected. Causal evaluation asks a different question: does the frozen decoder assign more probability to the gold answer after that evidence becomes native K/V? We intervene at both the selection and consumption boundaries. The no-memory condition supplies only the question. Oracle conditions force every 32-token parent intersecting an annotated evidence span. The direct-text control places the same evidence in the ordinary prompt. Four deterministic examples from each dataset make this a mechanism probe rather than an accuracy estimate.

Forced identities enter `prepare_selected_memory`, exactly where ordinary semantic routing ends. Both routes then share thresholding, whole-parent budgeting, native post-RoPE K/V, CPU-to-GPU transfer, GQA layout, source positions, additive masks, and Qwen's eager attention kernel. A selected parent identity is fixed across layers, but its tensor is not: layer ℓ receives $K_c^{(\ell)}, V_c^{(\ell)}$ captured from that layer's own Qwen projections and RoPE path. The 640-token bound reserves 128 direct tokens and permits up to 512 memory tokens per layer. Every oracle parent survives; no budget or native-limit violation occurs, and each reference-encoding call is at most 128 tokens. "Active K/V" is the sum of materialized physical tokens over consuming layers, not the number of distinct source tokens.

Generation-only F1/EM is too coarse for this small frozen checkpoint. The primary paired outcome is therefore teacher-forced mean gold-token log-probability,

$$\Delta_{c,i} = \frac{1}{|y_i|} \log p(y_i \mid q_i, c) - \frac{1}{|y_i|} \log p(y_i \mid q_i, \varnothing), \tag{10}$$

with first-token probability/rank, generated text, attention mass, and answer-position hidden state deltas retained in per-example traces. Four deterministic examples from each dataset are used. This is a mechanism probe, not an accuracy estimate.

HotpotQA supports the integration-depth hypothesis within a late band. The last-one, last-four, last-eight, and last-half interventions move mean log-probability by +0.071, +1.004, +2.311, and +3.641 nats/token. The last eight recover 49.9% of the direct-text effect while using 8/28 layers; the last half recover 78.6% at 1.75 times the last-eight materialization. Generated F1 does not change. The evidence therefore establishes causal likelihood use, not decoded QA success.

This depth dependence is plausible because a residual transformer accumulates context-conditioned updates,

$$h^{(\ell+1)} = h^{(\ell)} + \Delta h^{(\ell)}. \tag{11}$$

Memory introduced too late has few opportunities to affect that trajectory. Memory introduced at every layer can interfere before the local question representation has formed. The measured late band lies between those extremes, but the small probe does not establish a universal theory of transformer hierarchy.

QASPER separates perturbation from interpretable utility. It turns positive at the last eight layers (+0.347) but is only +0.072 at the last half, while direct evidence text is negative (−1.791).

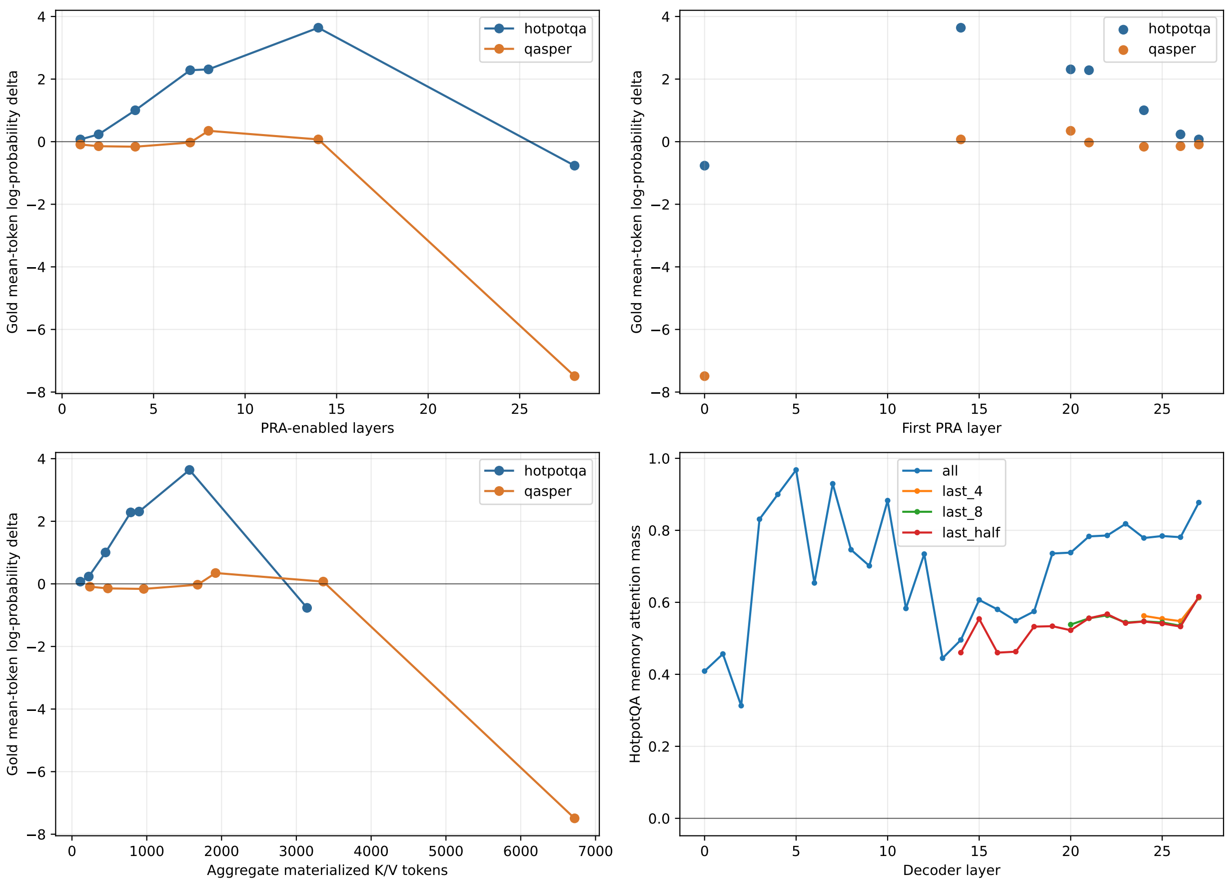

Figure 5: **Retrieved native K/V has a depth-dependent causal effect.** A contiguous late band approaches the direct-text benefit on HotpotQA, whereas all-layer sparse injection is harmful. The lower panels show that additional exposure also increases materialization cost.

Table 12: Oracle consumption-depth sweep. $\Delta \log p$ is paired mean gold-token log-probability relative to no memory. K/V is aggregate physical materialization across active layers. The key comparison is non-monotone: a late band helps on HotpotQA, whereas all-layer injection harms both datasets.

Dataset	Condition	Layers	$\Delta \log p$	F1	K/V tokens	TTFT	Generation
HotpotQA	none	0	.000	.056	0	.140	.737
HotpotQA	oracle, last 1	1	+.071	.056	112	.143	.733
HotpotQA	oracle, last 4	4	+1.004	.056	449	.154	.813
HotpotQA	oracle, last 8	8	+2.311	.056	898	.169	.877
HotpotQA	oracle, last half	14	+3.641	.056	1,572	.193	1.013
HotpotQA	oracle, all	28	-.766	.000	3,143	.245	1.382
HotpotQA	direct evidence text	0	+4.635	.125	0	.198	.986
QASPER	none	0	.000	.125	0	.135	.781
QASPER	oracle, last 1	1	-.091	.125	240	.137	.785
QASPER	oracle, last 4	4	-.161	.125	960	.149	.844
QASPER	oracle, last 8	8	+.347	.125	1,920	.163	.899
QASPER	oracle, last half	14	+.072	.100	3,360	.188	1.081
QASPER	oracle, all	28	-7.492	.000	6,720	.249	1.488
QASPER	direct evidence text	0	-1.791	.000	0	.416	1.090

That control failure prevents using this subset to grade PRA transport. Attention mass and hidden-state perturbations are retained in the artifact, but neither alone is evidence that a perturbation is useful.

A.1 Why oracle does not equal native context

The all-layer result could have been a bookkeeping error rather than a model effect, so we first reproduced the canonical cohort and audited the intervention end to end. Hotpot supporting-fact title and sentence IDs, and the corresponding QASPER evidence annotations, were resolved to exact source character spans, tokenizer spans, intersecting parent IDs, and physical materialized token positions. Selection used no answer text, answer token, or target likelihood. Every representable annotation was covered at every requested layer; all 28 Qwen layers were present; overlap deduplication introduced no repeated K/V; masks exposed every retained token; and no parent was rejected by the budget. A fresh replay of reference publication matched every cached post-RoPE K and native V exactly (maximum absolute error 0). The direct-text, last-14, and all-layer likelihood deltas reproduce the frozen artifact with maximum absolute difference 0. We therefore find no hidden oracle-path or protocol bug.

The native sufficiency ladder narrows the first hypothesis. On HotpotQA, exact annotated sentences gain +4.635, their parent paragraphs gain +5.574, and feasible full context gains +3.771 nats/token. Annotated evidence is therefore sufficient for this frozen model, although parent context supplies a useful bridge. The QASPER annotation and parent conditions are identical on this slice and both lower likelihood; full papers exceed the matched 2,048-token control. That cohort cannot distinguish annotation insufficiency from prompt incompatibility.

The representation comparison removes a second ambiguity. We align identical evidence BPE tokens between the direct prompt and bounded source publication. Late-band pre-RoPE K cosine is .940 on HotpotQA and .963 on QASPER; V cosine is .892 and .926. Corresponding relative errors are .339/.448 and .259/.349. Post-RoPE K cosine falls to .814/.907 because it also includes the different logical positions. The cache itself is exact: this discrepancy arises before materialization

Table 13: Oracle-gap audit on the canonical four-example-per-dataset cohort. E is annotated evidence, D other tokens in selected parents, and H the ordinary local/head support. K/V similarities average aligned evidence tokens over the last 14 layers. The counterfactual removes D from the existing logits and renormalizes; it does not change real materialization.

Diagnostic	HotpotQA	QASPER	Interpretation
Oracle evidence coverage	100%	100%	Annotation-to-parent selection is correct
Exact evidence $\Delta \log p$	+4.635	-1.791	Hotpot annotations suffice; QASPER lacks a positive text control
Parent context $\Delta \log p$	+5.574	-1.791	Bridge/parent context helps Hotpot
Late-band pre-RoPE K / V cosine	.940 / .892	.963 / .926	Source-only and question-conditioned representations differ
Late-band $M_E/M_D/M_H$	.220/.310/.469	.565/.027/.408	Co-materialized non-evidence support is substantial only on Hotpot
M_E after offline removal of D	.281	.579	Renormalization recovers .061 versus .014 evidence mass
Final hidden relative L2, last-14 / all	.541/.869	.602/1.245	All-layer replay accumulates greater internal divergence

because direct evidence is jointly contextualized with the question, whereas a reference block is encoded independently.

Softmax support contributes on HotpotQA but is not the complete explanation. In the last 14 layers, D receives .310 of normalized attention support. Removing it only in an offline counterfactual raises M_E from .220 to .281 and lowers entropy from 2.045 to 1.711. Under all-layer replay, D rises to .423 and counterfactual M_E rises from .283 to .438. Yet native direct evidence succeeds with only .120 evidence mass, below either PRA condition. On QASPER, D is only .027 in the late band and .015 across all layers. Raw evidence mass is therefore neither necessary nor sufficient for useful integration.

The residual trace provides the common proximate signal. Final answer-position relative L2 grows from .541 to .869 on HotpotQA and from .602 to 1.245 on QASPER when exposure expands from the last 14 to all 28 layers. Mean layerwise divergence grows from .104 to .642 and from .145 to .759. Direct context participates in the ordinary sequence from the first layer; source-only memory is instead introduced as parallel attention support. Repeating that intervention before a stable question representation forms compounds the contextual mismatch.

A one-example-per-dataset encoding-context sweep holds the final 32-token parent spans fixed. Hotpot last-14 benefit rises from +2.985 at 32 tokens to +3.548 at 256 and 512. QASPER falls from +0.908 to +0.230 while its V cosine improves from .876 to .960. Longer source context can matter, but representation cosine is not a monotone proxy for causal utility; this sweep is a mechanism diagnostic, not an estimate over the cohort.

The audit supports multiple effects rather than a single-cause account: bounded source encoding changes the payload representation; Hotpot parents add softmax support that dilutes evidence; and all-layer injection compounds residual-path divergence. Paper 2 diagnoses these factors without altering the selected payload. Evidence-centered and adaptive materialization remain the scope of Paper 3.

A.2 Placement is not interchangeable

The depth result justifies one placement follow-up. Table 14 fixes either four or seven consuming layers and leaves oracle parent identities, positions, prompts, and per-layer payload sizes unchanged.

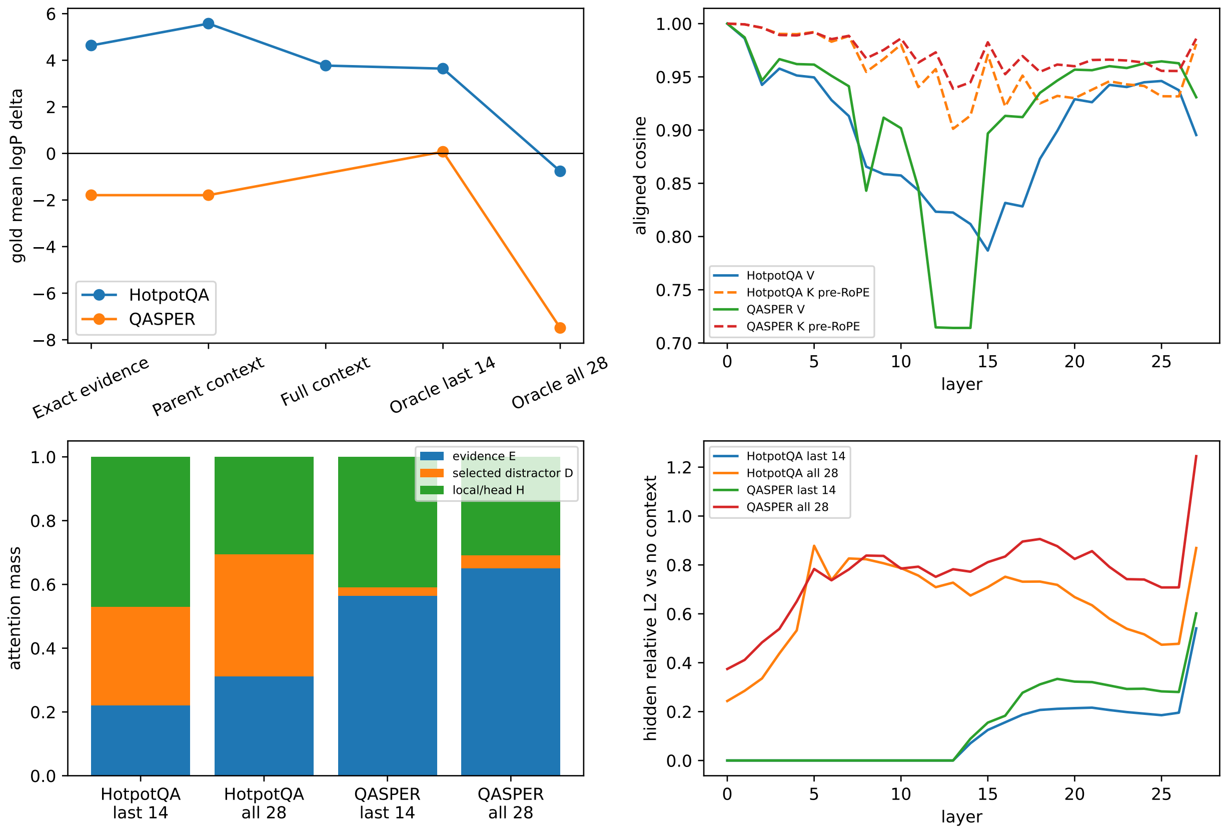

Figure 6: **The oracle gap combines representation, support, and depth effects.** The native context ladder establishes Hotpot annotation sufficiency; aligned K/V differ after contextual encoding; co-materialized non-evidence tokens absorb Hotpot attention; and all-layer replay departs much further from the no-memory residual trajectory than last-14 replay.

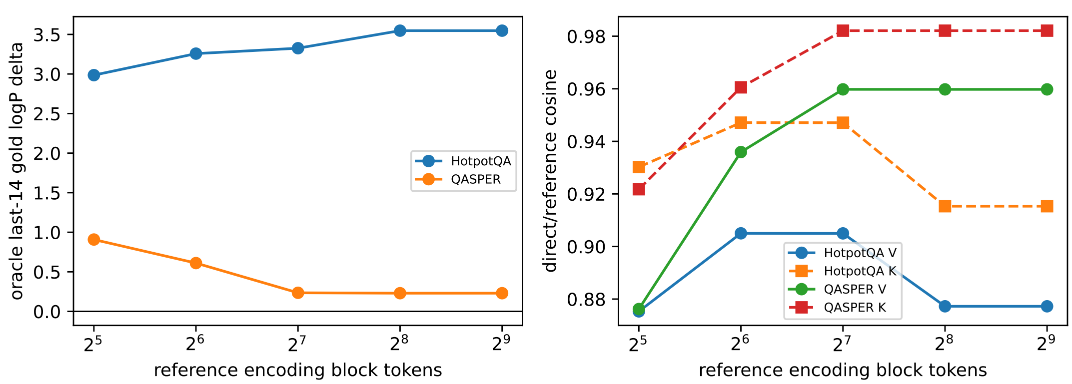

Figure 7: **Reference encoding context has dataset-dependent effects at fixed materialization.** Utility and aligned representation similarity do not move monotonically together. Each curve contains one fixed example and should be read as a sensitivity trace.

Table 14: Oracle placement at fixed memory identities. Equal-count rows have equal aggregate K/V cost, isolating where memory enters the residual trajectory.

Placement	Layers	First layer	$\Delta \log p$	F1
Early contiguous	4	0	-10.573	.000
Middle contiguous	4	12	-.556	.090
Evenly spaced	4	0	-10.398	.000
Late contiguous	4	24	+.421	.090
Every fourth	7	0	-6.090	.000
Late contiguous quarter	7	21	+1.128	.090

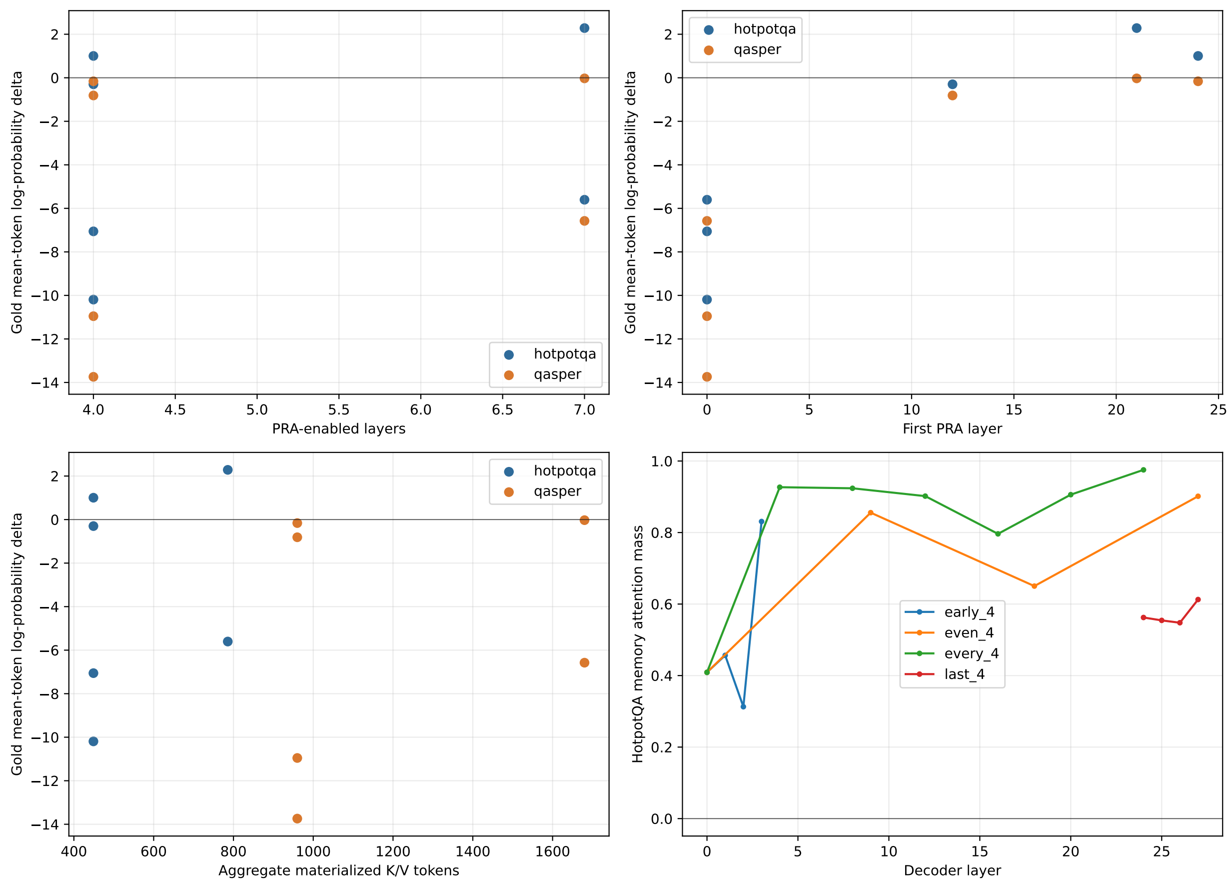

Figure 8: **A contiguous late band is not interchangeable with earlier or sparse placement.** At matched parent identities and per-layer payload, earlier schedules reverse the likelihood effect despite equal K/V cost at equal layer counts.

This rejects both proposed shortcuts. Earlier insertion does not help by leaving more downstream depth, and every-fourth insertion does not approximate dense late consumption. The likely mechanism is a layer-specific contextualization/topology mismatch: independently encoded reference states are least compatible with early residual computation, whereas a contiguous upper band can repeatedly integrate them after the local question has become semantically formed. This interpretation is mechanistic, not yet a general theorem about decoder depth. All-layer replay is especially informative: more attention opportunities can make the answer worse. The operating point is therefore a band, not “as many PRA layers as possible.”

A.3 Route once, consume with layer-native payloads

The routed follow-up scores the question once at layer 27, selects three parent IDs, and reuses those identities across either the last 8 or last 14 layers. Every layer resolves the ID to its own post-RoPE K/V; no tensor crosses layer boundaries. Table 15 reports the complete pre-top-k evidence-identity ranking and the aggregate consumption cost.

Table 15: Combined route-once result over eight examples. Recall is evidence-identity recall, the fraction of mapped evidence-parent identities recovered; f_{80} and f_{90} are the first measured parent fractions reaching those levels.

Router	Consumption	ID R@5%	ID R@10%	ID R@20%	ID R@30%	f_{80}	f_{90}	$\Delta \log p$	F1	K/V
Learned margin	last 8	.264	.403	.521	.639	.502	1.000	+.227	.090	768
Learned margin	last half	.264	.403	.521	.639	.502	1.000	+.374	.028	1,344
Raw hidden-state cosine	last 8	.092	.224	.414	.573	1.000	1.000	+.150	.090	768
Raw hidden-state cosine	last half	.092	.224	.414	.573	1.000	1.000	+.568	.090	1,344

The learned router gives the stronger sparse ranking and reaches evidence in 5/8 selected top-3 sets. Last-eight consumption preserves the no-memory F1 and moves likelihood by +0.227; last-half consumption moves it by +0.374 but lowers F1 to .028. Raw cosine has weaker sparse recall yet a larger last-half likelihood shift. This is not a contradiction: a selected non-annotated parent can still change the answer distribution, and a likelihood movement need not cross the greedy decoding boundary. The learned last-eight condition is the conservative quality/cost operating point. It materializes 768 K/V tokens across layers, increases TTFT from .141 to .167 seconds and generation from .759 to .907 seconds, and adds one .149-second query encoding plus .021-second exact route. Its layer-27 learned index averages 45,184 bytes; storing projected gists for all 28 instrumented layers averages 1.27 MB, versus 322.7 MB of resident detail K/V. These eager single-example timings characterize this implementation only.

A.4 Retrieval, causal influence, and decoding are distinct

The earlier single-layer `#__head` run established bounded displacement but generated no correct answer. We repeat that probe with the learned router at layer 27 and the selected late-eight consumption band. The 416-token logical prompt displaces 288 tokens, retains a 128-token direct tail with positions 288–415, and indexes nine 32-token parents. The router ranks the evidence parent first and selects it with two neighboring parents. Each layer 20–27 then consumes its own 96-token native K/V payload.

Table 16: Bounded implicit-head confirmation. The router ranks the evidence first and selected memory strongly improves the gold distribution, but the greedy answer remains wrong. Rank is the gold first-token vocabulary rank; K/V is aggregate physical materialization.

Condition	Evidence rank	$\Delta \log p$	First-token rank	K/V	F1	Viol.
No memory	–	.000	94,106	0	.000	0
Learned route, last 8	1	+5.469	1,351	768	.000	0

The intervention raises the two-token answer’s mean log-probability from -10.924 to -5.455 and its first-token probability from 3.79×10^{-9} to 1.95×10^{-5}. Greedy output still omits “cobalt.”

The maximum native operation remains 128 tokens and no violation occurs. Thus one bounded run demonstrates successful discovery, layer-native transport, and strong causal likelihood use together. It also shows why they must be measured separately from decoded task success: a five-order-of-magnitude first-token probability increase need not cross the greedy decision boundary.

A.5 Integration ceiling and routed adaptation across domains

The depth sweep identifies layers 14–27 as Qwen's strongest measured memory-consumption band. We therefore rerun the complete memory-use ladder at that depth instead of extrapolating from the preliminary late-four study. This experiment fixes the backbone, router, selected native K/V, prompt, and attention budget. It varies only M_ψ, the PRA-active transformation of the memory effect. The residual alternative forms

$$d_l = y_l^{\text{mem}} - y_l^{\text{local}}, \qquad y_l^{\text{res}} = y_l^{\text{mem}} + W_{\text{up},l}\,\text{SiLU}(W_{\text{down},l} d_l + b_l), \tag{12}$$

where the bottleneck width is 16, 32, or 64. The LoRA alternative modifies the native output projection only on a PRA-active call:

$$y_l^{\text{LoRA}} = W_{O,l} z_l + \frac{\alpha}{r} B_l A_l z_l, \qquad r \in \{4, 8\}, \quad \alpha = r. \tag{13}$$

Both up projections start at zero. The required combination adds both corrections after the same native memory attention. Every alternative is structurally bypassed when PRA is inactive. Thus logits and greedy generation remain bit-exact to vanilla Qwen without relying on a regularizer to approximate parity.

The protocol trains on 12 HotpotQA identities for 32 oracle-memory updates, selects width and rank on four disjoint HotpotQA plus four disjoint QASPER identities, and tests on a further eight identities from each dataset. Seeds 11, 23, 37, 53, and 71 vary initialization and example order. Validation selects residual width 16 and LoRA rank 4; their combination is then trained without consulting test identities. The primary score is the gold sequence log-probability change from the same no-context prompt. Recovery is a ratio of matched cohort means, and is reported only when the direct or full-context denominator is positive and greater than 0.05.

Table 17: Decision-relevant rows from the identity-disjoint last-14 diagnostic ladder. Parameters count M_ψ only; every row also uses the fixed router. H/Q denote HotpotQA/QASPER and O/R oracle/routed selection. Entries are sequence $\Delta \log p$; Q F1/EM is routed. Complete widths, ranks, and controls remain in the released artifact.

Method	M_ψ	H O	H R	Q O	Q R	Q F1/EM
Frozen PRA	0	+11.212	+1.901	+1.444	+1.377	.073/.000
Residual 16	458,976	**+14.478**	**+1.395**	+5.067	+4.942	**.205**/.000
Output LoRA $r = 8$	344,064	+11.057	-6.979	**+5.177**	**+5.589**	.135/.000
Residual 16 + LoRA 4	631,008	+12.035	-5.051	+5.140	+5.583	.171/.000

QASPER gives the clearest positive adaptation result in this initial diagnostic ladder. Every trained conditional alternative improves routed sequence likelihood over frozen PRA. Residual 16 raises the routed gain from +1.377 to +4.942, the preliminary LoRA rank-8 row reaches +5.589, and residual 16 reaches routed F1 .205. The later identity-disjoint held-out sweep in Section A.6 selects rank 32 as the final LoRA configuration. The improvement survives router selection rather than appearing only under oracle memory. This is consistent with QASPER being the router's training domain, although the experiment does not isolate domain match as the cause. Direct evidence itself changes sequence likelihood by -1.182 on this cohort, and no full source fits the 2,048-token control limit, so recovery ratios are undefined and the result supports routed likelihood integration rather than a context-recovery or decoded-QA claim.

The larger cohort rejects complementarity. On HotpotQA oracle replay, residual 16 exceeds frozen PRA by 3.266 ± 0.298 sequence log-probability and does so for all five seeds. It recovers 90.8% of the direct-evidence benefit and 107.3% of the feasible full-context benefit while materializing 11.39% of source K/V. Its oracle F1/EM is .314/.250. Yet under learned routing, residual 16 gains only 1.395 versus frozen PRA's 1.901 and recovers 8.8% rather than 11.9% of direct evidence. The paired difference from frozen is $-.506 \pm .523$ and changes direction across seeds.

LoRA and the combination fail more sharply on routed HotpotQA. The combination trails residual 16 by -6.446 ± 0.551 in every seed and trails LoRA by $-.881 \pm 1.280$ without a consistent seed direction. Oracle residual and LoRA therefore do not repair independent mismatches that can be profitably stacked. Training on clean oracle memory instead makes the larger conditional updates brittle to routing errors. The final end-to-end architecture consequently retains frozen last-14 PRA plus R_ϕ; residual 16 is an integration-ceiling instrument, not a default product component.

A plausible interpretation is selection-distribution shift. Adapter training presents clean, annotated memory, whereas deployment presents the router's selected memory and its characteristic omissions and distractors:

$$P_{\text{train}}(M) \approx P_{\text{oracle}}(M) \neq P_{\text{router}}(M) \approx P_{\text{deploy}}(M). \tag{14}$$

The oracle-to-routed HotpotQA reversal is consistent with this mismatch but does not prove it. It shows that the memory consumer and router cannot be assumed to optimize independently across domains. Future multi-hop experiments should train and validate memory use on the iterative router's actual selections and errors, not only on one-shot gold memory.

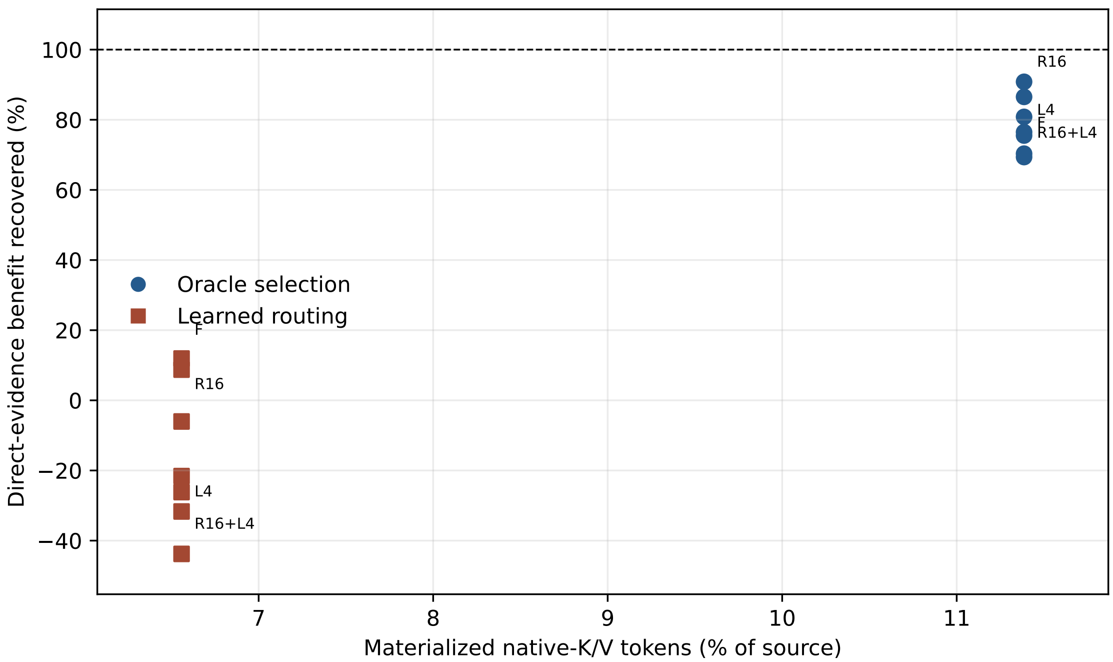

Figure 9: HotpotQA direct-evidence recovery versus physical native-K/V materialization. F, R16, L4, and R16+L4 label the frozen, residual-16, LoRA-4, and combined controls; unlabeled points are the remaining registered widths/ranks. All methods share a fixed memory fraction within a selection mode. Oracle residual 16 raises the ceiling, while every learned adaptation lowers the routed frontier relative to frozen PRA.

Table 18: Physical memory economics, averaged over the test identities. Active transfer is the selected native K/V copied for all 14 consuming layers. Peak GPU allocation includes the model and is not an incremental K/V measurement. The routing-index ratio compares compact index bytes with the complete CPU detail-K/V bank.

Dataset	Selection	Chunks %	K/V tokens	K/V %	Index/detail %	CPU detail MiB	Transfer MiB
HotpotQA	oracle	11.29	167.1	11.39	.0282	82.35	9.14
HotpotQA	routed	6.60	94.9	6.56	.0282	82.35	5.19
QASPER	oracle	4.68	192.0	4.70	.0280	245.54	10.50
QASPER	routed	2.50	96.0	2.51	.0280	245.54	5.25

The mean peak GPU allocation is 1.17–1.18 GiB for these rows. The selected HotpotQA product point transfers 5.19 MiB per example and materializes 6.56% of source K/V; frozen routed PRA recovers 11.9% of direct-evidence and 14.1% of full-context sequence-likelihood benefit there. Requested chunk fraction, realized token fraction, index overhead, off-device detail storage, and active transfer are deliberately reported separately.

Table 19: HotpotQA rank and decoding diagnostics on the larger cohort. Probability, rank, margin, sequence likelihood, and lexical answer quality are reported separately. Negative margin means the gold first token remains below the model's top token.

Condition	Gold prob.	Gold rank	Margin	Sequence logP	F1	EM
No context	.0616	55.6	-6.513	-20.192	.025	.000
Frozen PRA, routed	.0316	34.1	-6.326	-18.291	.000	.000
Residual 16, routed	.1185	121.9	-6.659	-18.797	.028	.000
LoRA 4, routed	.1354	274.9	-8.962	-24.363	.049	.000
Residual 16 + LoRA 4, routed	.1312	284.0	-9.232	-25.243	.022	.000
Residual 16, oracle ceiling	.4094	22.8	-1.780	-5.714	.314	.250
Direct evidence text	.1973	12.4	-3.112	-4.251	.257	.125
Full original context	.3018	16.3	-3.176	-6.704	.275	.250

Memory-use quality cannot be summarized by a single scalar: an intervention can raise the target token's absolute probability while worsening its rank against competing tokens or reducing the gold sequence likelihood. Frozen routed PRA improves sequence likelihood and mean first-token rank but does not improve F1/EM. Oracle residual 16 reaches EM .25 and raises the gold probability to .409, yet the mean rank is still 22.8 and the margin remains negative. Reliable decoded QA is not closed by this adaptation ladder.

A.6 A bounded LoRA sweep strengthens QASPER integration and exposes a multi-hop gap

The first ladder leaves a narrow ambiguity: perhaps ranks 4 and 8, 32 updates, or the baseline learning rate underfit the conditional output map. We test that explanation without changing the last-14 placement, native K/V, router, attention budget, or oracle-memory training objective. Stage A screens ranks 4, 8, 16, and 32 for 32 and 64 updates at learning rate 10^{-3}. Stage B expands the three best ranks to 64 and 128 updates at 0.5×, 1×, and 2× that rate. The score is equal-weight mean oracle sequence $\Delta \log p$ over four held-out HotpotQA and four held-out QASPER identities. Three rank-diverse finalists then run over seeds 11, 23, 37, 53, and 71. Test identities remain unopened until this choice is frozen.

The validation-only Pareto rule selects rank 32, 64 updates, and learning rate 10^{-3}: 1,376,256 parameters, or 0.231% of Qwen3-0.6B. Rank 32 improves from 11.704 at 32 updates to 12.484 at 64 in the single-seed screen, then falls to 12.096 at 128. At the baseline rate, ranks 4 and 8 instead decline when extended from 32 to 64 updates. Longer optimization is thus selectively useful rather than a general cure for under-convergence. Rank 64 is not expanded: rank 32 already fails the learned-routing product gate, so these data do not establish saturation beyond the tested rank.

Table 20: Five-seed untouched-test result for the bounded conditional-LoRA sweep. H and Q denote HotpotQA and QASPER; O/R denote oracle/routed selection. ρ_D and ρ_F are cohort recovery against direct evidence and feasible full context. F1 is oracle/routed. QASPER recovery is undefined because direct evidence lowers likelihood on this cohort.

Method	M_ψ	% base	H O	H R	H ρ_D O/R	H ρ_F O/R	H F1 O/R	Q $\Delta \log p$ O/R
Frozen PRA	0	.000	+11.212	+1.901	70.3/11.9	83.1/14.1	.102/.000	+1.444/+1.377
LoRA $r = 32$	1,376,256	.231	**+13.375**	-8.640	**83.9**/-54.2	**99.2**/-64.1	**.310**/.022	+4.641/+4.967
Residual 32 + LoRA 32	2,294,208	.385	+13.677	-8.057	85.8/-50.5	101.4/-59.7	.268/.048	**+5.062/+5.363**

On QASPER, the selected rank-32 LoRA raises routed sequence-likelihood gain from +1.377 to +4.967, a $+3.590 \pm 1.246$ paired improvement. The residual-plus-LoRA control reaches +5.363. These untouched-test results strengthen the positive in-domain finding from the smaller ladder.

They do not yet establish decoded QA or a recovery fraction because QASPER has no positive direct-text denominator on this cohort.

HotpotQA exposes the complementary robustness boundary. Rank-32 LoRA improves oracle memory by 2.164 ± 1.425 nats relative to frozen PRA in all five seeds, but its paired routed effect is -10.541 ± 4.841, also in the same direction in all five seeds. Mean routed gold rank worsens from 34.1 to 124.9 and EM remains zero. The contrast is dataset-conditional: adaptation improves routed QASPER likelihood but not one-shot routed HotpotQA memory.

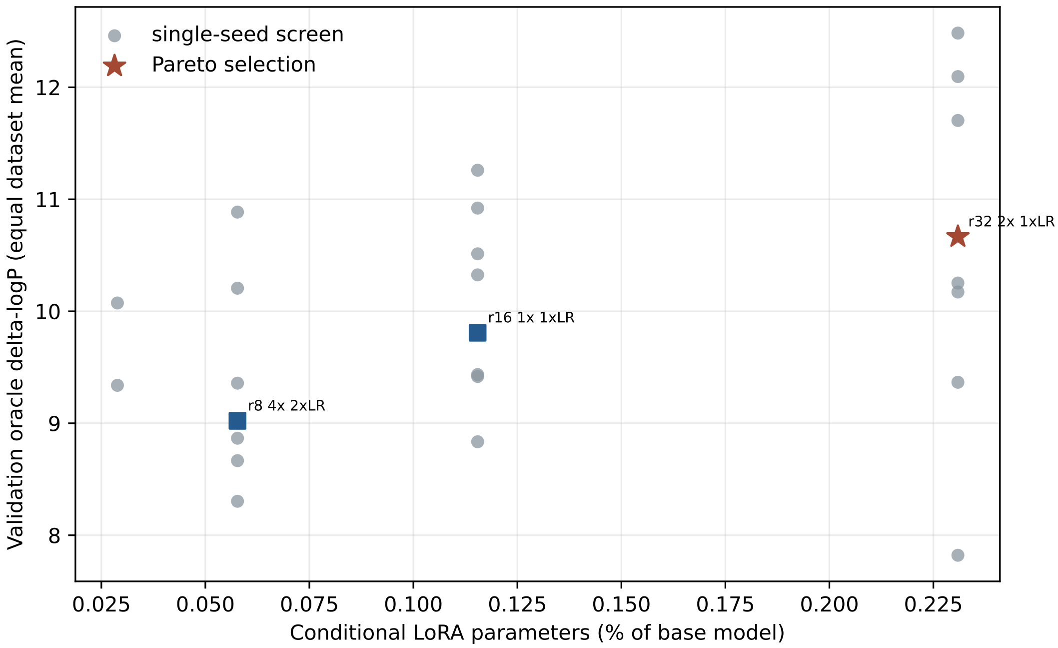

Figure 10: Held-out oracle sequence benefit versus conditional-LoRA parameter fraction. The staged screen identifies rank 32 at 64 updates as the validation Pareto choice; on untouched QASPER it raises routed sequence benefit from +1.377 to +4.967. Its separate routed HotpotQA result keeps it research-only pending iterative multi-hop evaluation.

One preregistered combination adds residual width 32 to the selected LoRA. Its combo-minus-LoRA effect changes direction across seeds for both datasets and both selection modes; the HotpotQA routed mean is only $+0.583 \pm 6.697$. We reject the combination and retain frozen last-14 PRA plus the learned router as the SDK default. The rank-32 weights are released only as a research opt-in. All 43 candidate PRA-off checks are exact, and no native-operation-limit violation occurs.

A.7 PRA-conditional adaptation preserves ordinary language

Memory adaptation is useful only if it does not silently rewrite the model's ordinary language behavior. The PRA-conditional gate, residual, and output-LoRA paths are absent when no external memory is selected. On 1,024 fixed WikiText-2 validation tokens, every conditional variant in the matched global-readout control retains the frozen loss of 3.8811 exactly. Their no-PRA QA log-probability changes are also zero. This is a positive architectural result: memory-specific capacity can improve routed QASPER likelihood while preserving the original no-memory model path.

Table 21: Ordinary-language preservation control. WikiText-2 values are five-seed means on the same 1,024 validation tokens. Conditional methods are structurally bypassed without PRA memory; global readout controls are not.

Method	WikiText-2 loss	Δ loss from frozen
Frozen / PRA-conditional gate, residual, or output LoRA	3.8811	0.0000
Global LM-head LoRA $r = 8$	3.8972	+0.0161
Full global readout controls	3.9749–3.9750	+0.0938

The control also explains why global readout tuning is not the next adaptation step. It neither improves the memory-specific increment nor preserves ordinary-language loss. PRA-conditional adaptation keeps the stronger ownership boundary: it can specialize memory use without changing the function evaluated when memory is absent.

A.8 Behavioral Utility Beyond Likelihood and F1

Distributional utility (logits, probability, rank, and likelihood), reference fidelity (F1 and EM), decision-specific utility (polarity accuracy and margin), and behavioral utility (blind semantic equivalence, relative quality, and independent validity) measure different properties. Weak lexical overlap therefore cannot establish that an answer is useless, just as a likelihood gain cannot establish that a generated answer is useful. To close this measurement gap, two external judges—GPT-5.6 Sol and Claude Sonnet 5—scored a frozen set of 152 underlying comparisons presented as 304 opaque A/B-reversed items. Full protocol, hash, validation, and judge-specific details appear in Appendix I.

Table 22: Convergence of QASPER evidence for residual-16 adaptation. Preference rates are mutually exclusive positive/negative relative-quality categories; the remaining comparisons receive zero relative-quality score.

Measure	Frozen PRA	Adapted PRA	Interpretation
Routed sequence $\Delta \log p$	+1.377	+4.942 residual-16	Same-condition probability trajectory improves
Routed F1 / EM	.073 / .000	.205 / .000 residual-16	Lexical recovery improves but remains limited
Blind relative-quality preference	2.5%	22.5% residual-16	Both judges favor adaptation more often than frozen PRA
Cross-judge agreement	–	$r = .991$ Eq.; $r = .995$ quality	QASPER judgments are highly consistent by dimension

The likelihood and behavioral instruments converge on QASPER: residual-16 improves routed sequence likelihood and is preferred behaviorally more often than frozen PRA despite weak F1 and zero EM. Both judges are perfectly invariant to exact A/B reversal after orientation, so answer position does not explain the preference. This does not make the judges ground truth. Across all 152 pairs their semantic-equivalence scores correlate strongly ($r = .845$), while relative-quality agreement is only modest ($r = .375$); Claude Sonnet 5 also under-scores literal identical-answer calibration. We therefore keep judge-specific estimates and do not pool them.

A.9 A 32-token follow-up isolates QASPER polarity

The frozen behavioral package uses an eight-token decoding budget. That budget made the blind comparison reproducible, but it often stopped before an answer reached punctuation or EOS. We do not alter or rescore those 152 comparisons. Instead, a separate diagnostic replays the same test identities with a 32-token budget while leaving model, router, selected memory, prompts, and checkpoints unchanged. Increasing the budget removes most incomplete-output confounding; the matched completion table and full protocol are in Appendix H.

Longer decoding removes most incomplete endings for frozen PRA and residual 16, but it does not remove the QASPER decision error. All frozen routed outputs begin with a valid yes/no form, yet only 25.0% have the correct polarity. Oracle memory raises this to 62.5%. The existing residual-16 adapter, trained on clean HotpotQA memory, reaches 72.5% across five seeds, raises answer containment from .500 to .825, improves mean first-token rank from 25.25 to 4.43, and ends with EOS in 95.0% of trials. The improvement therefore reflects more than completion alone. The routed-to-oracle comparison shows that better evidence selection substantially improves the frozen consumer's final binary decision; it is not an independent additive estimate of router causality. Residual 16 then exceeds the oracle-frozen result, indicating that conditional memory-use adaptation can contribute further decision quality. Both comparisons use eight fixed test identities, so their value is mechanistic rather than benchmark-level.

Table 23: QASPER decision and completion diagnostics at 32 generated tokens. R16-Q is trained for 32 updates on the shipped router's selected QASPER memory. Adapter rows are optimization means over five seeds on the same eight identities; frozen rows are deterministic replays.

Condition	Format	Polarity	Contain.	F1	m_{pol}	EOS
Frozen routed	1.000	.250	.500	.048	-.078	.875
Frozen oracle	1.000	.625	.625	.061	.260	.625
Residual 16	1.000	$.725 \pm .056$	.825	.152	.243	.950
R16-Q routed-trained	1.000	$.550 \pm .112$	.550	.511	.143	1.000

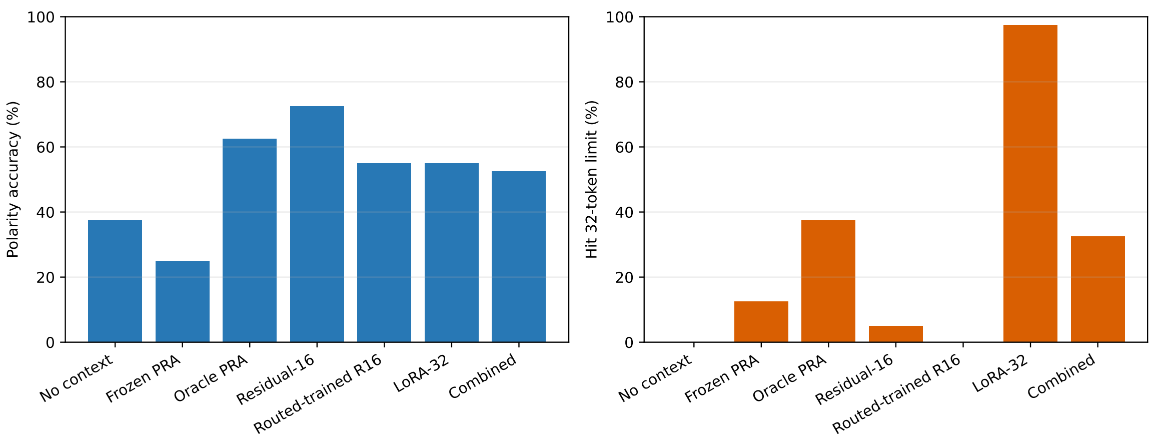

Figure 11: QASPER polarity accuracy and 32-token length termination. The residual adapter improves the evidence-conditioned decision while retaining natural completion. Rank-32 LoRA has a positive teacher-forced polarity margin but reaches the generation cap in 97.5% of trials; probability improvement alone therefore does not establish decoded utility.

Three small controls do not close the remaining gap. Restricting the first token to yes/no retains 25.0% accuracy. A two-parameter affine calibration fitted on 12 disjoint QASPER identities reaches 50.0% on the eight test identities. A top-one–top-two router-margin gate selects full memory on validation and reproduces frozen PRA at 25.0%. Training residual 16 directly on routed QASPER memory produces concise answers—100% EOS, mean first-token rank 1.45, and F1 .511—but polarity is only 55.0%, below the oracle-trained residual. This mixed result is consistent with small-data readout specialization rather than a general deployment-distribution solution. The experiment stops here: residual 16 remains the strongest balanced QASPER condition, and frozen PRA plus the router remains the generic SDK default.

The decomposition separates two adaptation sub-objectives. *Decision formation* asks whether retrieved evidence produces the task-critical branch—yes versus no here. *Answer realization* asks whether the model can emit a fluent answer-shaped continuation and terminate. **Learning to produce an answer-shaped sequence is not the same as learning the evidence-conditioned decision that sequence should express.** The routed-trained residual learns realization strongly—F1 .511, EOS 1.000, first-token rank 1.45, and sequence log probability $-.961$—yet polarity is only 55.0%. The control exposes a broader objective mismatch: teacher-forced sequence optimization can strongly improve answer realization while leaving a sparse task-critical decision weak. This observation is not specific to PRA and merits separate study; it does not imply that cross-entropy is generally inadequate.

We tested whether explicit binary-decision weighting repairs that gap while holding the model, router, references, residual width, memory budget, optimizer, updates, generation length, and seeds fixed:

$$\mathcal{L} = \mathcal{L}_{\text{seq}} + \lambda \mathcal{L}_{\text{polarity}}. \tag{15}$$

Validation on four disjoint identities selects $\lambda = 2$, but untouched-test polarity falls from the exact sequence-only baseline of 55.0% to 40.0%, below the 62.5% test majority baseline. Thus this small-data result provides no evidence that simple first-token reweighting improves out-of-sample polarity. The $\lambda = 0$ rerun exactly reproduces all 40 prior seed–item rows across generated text and every reported metric. Full sweep, seed, class-balance, margin, no-memory, oracle-memory, and WikiText receipts appear in Appendix H and the released artifact.

For HotpotQA, the primary taxonomy over the five-seed residual condition contains 17 relation-near misses, five unchanged answers, five correct answers, five prompt repetitions, four unrelated outputs, three length terminations, and one wrong related entity. **The dominant HotpotQA error is not random generation but failure to complete a relevant relation to its endpoint.** The one-shot path often reaches $q \to A$ but lacks explicit closure to $q \to A \to B$. A reproducible lexical relation-distance audit is exported with the artifact, but it is heuristic rather than a dataset-native hop label. These errors support iterative relation closure in Paper 2.5; they do not motivate further one-shot tuning here.

HotpotQA does not reproduce the QASPER behavioral result: the judges give opposite mean directions for residual-16 adaptation. Together with routed adapter brittleness, the likelihood–decoding disconnect, and incomplete one-shot multi-passage discovery, this evaluator dependence motivates iterative evidence gathering in Paper 2.5 rather than more one-shot top-k tuning. Adapter choice also depends on evaluation level: the configuration that maximizes teacher-forced likelihood need not be the one most consistently preferred behaviorally.

Decision. PRA-specific capacity totals 262,144 parameters (0.044% of Qwen) in the selected routed system. The oracle residual probe adds 458,976 parameters, bringing the diagnostic total to 721,120 (0.121%); the rejected combination would total 893,152 (0.150%) and occupies 2.43 MiB on disk. The broader follow-up selects a 1,376,256-parameter rank-32 LoRA by held-out oracle likelihood and raises routed QASPER sequence benefit to +4.967. It remains a research opt-in rather than the SDK default because its one-shot routed HotpotQA change is −8.640 nats. PRA-off logits and generation are exact for every run, WikiText-2 confirms ordinary-language preservation for the tested conditional path, and native-limit violations are zero. These results freeze Paper 2's positive one-shot transport, discovery, integration, preservation, and QASPER behavioral-adaptation claims. The behavioral qualification applies to residual-16 decoded generations and does not validate the rank-32 checkpoint. HotpotQA is a multi-hop stress and causal probe here, not the final architecture: iterative evidence gathering, in which retrieved gists refine subsequent retrieval, is deferred to Paper 2.5.

B Cross-Family Validation Details

One PRA execution contract spans decoder families with materially different attention organization. We distinguish three claims. *Execution portability* means that the shared core and a thin family adapter preserve the host model's measured native semantics. *Routing portability* means that the same learned-router interface runs across families; its quality may still depend on model and domain. *Functional portability* means that routed native K/V measurably changes the frozen prediction. Qwen3-0.6B, the public SmolLM2-135M Llama-family control, and official Gemma 3 1B exercise the execution and routing interfaces; Qwen and Gemma also have functional causal probes. SmolLM2 is not a Meta Llama result; the official Meta Llama 3.2-1B checkpoint remains unmeasured.

All three routers use frozen late-layer attention-input states, one mean per 32-token parent, 128-dimensional asymmetric projections, 512 optimization steps, QASPER training, and seeds 11, 23, 37, 53, and 71. Their evidence-identity R@20% values are 0.454 ± 0.007, 0.447 ± 0.051, and 0.432 ± 0.025, respectively. The execution abstraction transfers across families; routing quality remains model- and domain-dependent.

Table 24: **Matched QASPER sparse routing is of comparable order across the three measured families, with model-dependent variance and no claim of universal routing quality.** "Exact" lists the strongest disabled-path quantities checked at the named checkpoint. QASPER R@20% is five-seed evidence-identity recall. Causal $\Delta \log p$ values use separate eight-example probes and are not directly comparable in magnitude. A dash means not measured.

Property	Qwen3-0.6B	SmolLM2-135M	Gemma3-1B-IT
Checkpoint tested	Official pinned	Public pinned Llama-family control	Official pinned
Exact PRA-off parity	Logits, hidden, greedy; cache shapes	Logits, hidden, greedy; cache shapes	Logits, hidden, greedy; cache tensors
Native K/V and cache policy	GQA, native cache	GQA, native cache	MQA, HybridCache
Host attention pattern	Full	Full	22 sliding + 4 global
PRA placement consumed	Late 8	Late 8	Global 17 and 23
QASPER identity R@20%	$.454 \pm .007$	$.447 \pm .051$	$.432 \pm .025$
Routed causal $\Delta \log p$	+.227	–	+.135
Zero native-limit violations	Yes	Yes	Yes
Wheel / public API smoke	Yes	Yes	Yes

Gemma is the structurally distinct test. Its official revision has global-capable layers 5, 11, 17, and 23, but the measured routed configuration enables and actually consumes selected memory only at layers 17 and 23; all 22 sliding-window layers remain native. On QASPER, its five-seed identity R@5/10/20/30% curve is .251/.326/.432/.514. The same router transfers more weakly to

HotpotQA, with $\text{AUC}_{0:30} = .196 \pm .014$, so the curve is evidence for a portable routing interface rather than equal routing quality.

Gemma therefore provides an architecture-informed memory-placement policy: PRA is admitted through the checkpoint's native global-attention structure while local sliding-window computation remains unchanged. This policy is consistent with the late-layer Qwen result, but the measured families do not establish a universal placement rule.

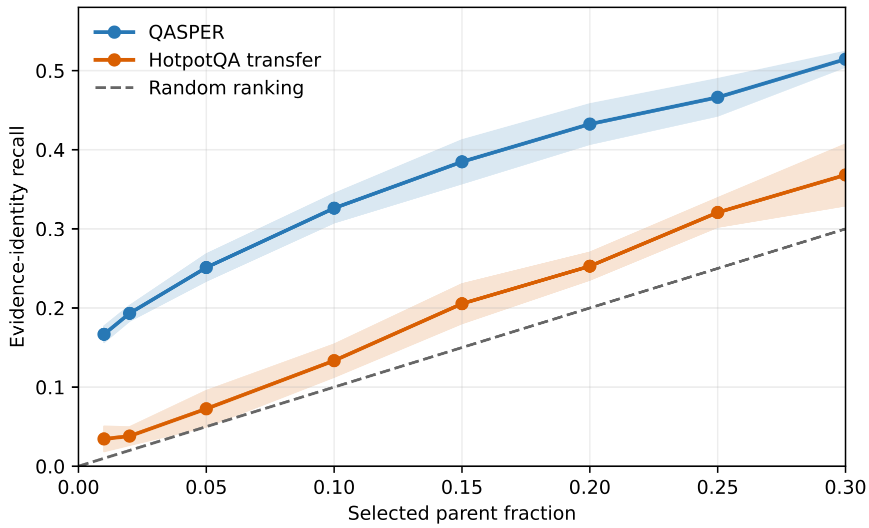

Figure 12: **Gemma routing quality is sparse-budget and domain dependent.** Five-seed evidence-identity recall is useful on QASPER, its training domain, and transfers more weakly to HotpotQA. Neither curve reaches 0.80 before exhaustive selection.

On eight deterministic causal-probe examples, routed Gemma memory enters layers 17 and 23 and raises mean gold-token log-probability by 0.135 nats while placing 0.060 attention mass on external memory. F1 remains 0.125, exactly equal to no memory. Forcing annotated parents into only the final global layer changes likelihood by -0.016 nats; direct evidence text changes it by $+3.765$ nats and reaches .536 F1. The measured chain is therefore routed memory $\rightarrow$ selected native K/V $\rightarrow$ positive gold-likelihood effect $\rightarrow$ no greedy-F1 change. It is a causal mechanism probe, not an accuracy estimate or evidence that Gemma+PRA outperforms Gemma.

B.1 Distributional, Lexical, and Behavioral Utility

Distributional utility is measured by logits, probability, rank, and sequence likelihood; *reference or lexical fidelity* by F1 and EM; *decision-specific utility* by polarity accuracy and polarity margin; and *behavioral or semantic utility* by blind equivalence, relative quality, and independent validity. No single scalar characterizes memory use. Qwen's bounded-head probe gains 5.469 nats and moves first-token rank from 94,106 to 1,351 while retaining the wrong greedy answer; Gemma gains 0.135 nats with unchanged F1. Conversely, weak QASPER F1 does not erase the independently judged residual-16 preference. These are different properties, not evidence that one metric is wrong.

C Product and Systems Details

The validated path is packaged as `pra-hf`, with a stable Python API, CLI, serializable configuration, fraction-based selection, and Hugging Face-style router artifacts. Version `0.2.0rc1` builds as a wheel and is pinned by tag `paper2-hf-rc1`. A caller adds a reference and generates through the same model object:

```
from pra_hf import PRAConfig, PRAForCausalLM

model = PRAForCausalLM.from_pretrained(
    BASE_MODEL, routing_adapter=ROUTER,
    pra_config=PRAConfig(selected_fraction=0.20))
ref = model.add_reference_file("paper.txt")
result = model.generate(
    question,
    return_details=True,
)
model.remove_reference(ref)
```

The API also supports in-memory text, explicit URI/text pairs, chat formatting, `#__head` rollover, cache clearing, and per-request statistics. A router artifact contains projection weights, configuration, and a model card; it pins the base model revision, family, routing layer and representation, parent size, dataset splits, seeds, metrics, PRA version, and source revision without republishing base-model weights. Qwen's router has 262,144 parameters (0.044% of 596,049,920); SmolLM2's has 147,456 (0.110% of 134,515,008); Gemma's has 294,912 (0.0295% of 999,885,955). Each stores one 128-dimensional FP32 vector, 512 bytes, per parent. The repository's `pra-hf-demo/pra_hf_model_families.ipynb` exercises the same public reference lifecycle and model-family interface for Qwen, Llama-family, and Gemma users.

Memory accounting uses three denominators. *Routing-index bytes* are the compact vectors searched once per request. *Resident detail bytes* are all layer-native K/V retained on CPU. *Materialized bytes* are the selected K/V transferred for each consuming layer. A 20% requested parent fraction need not become 20% active K/V: whole-parent rounding and the hard token cap may reduce it. Conversely, replaying one selected identity through eight layers incurs eight layer-specific payloads even though semantic routing happened once.

Table 25: **The same product path runs on Qwen3-0.6B, SmolLM2-135M, and Gemma3-1B-IT with bounded materialization.** Four-example CUDA means on a GTX 950M are a systems demonstration, not a throughput or QA benchmark. Peak GPU allocation includes the full model.

Family	Evidence	F1	Requested	Materialized	Route ms	Peak GiB
Qwen3-0.6B	0.667	0.033	20.4%	6.61%	6.40	1.126
SmolLM2-135M	0.750	0.028	21.0%	6.25%	32.74	0.261
Gemma3-1B-IT	0.708	0.417	20.6%	6.48%	14.91	1.935

Qwen averages a 45.9 KiB routing index, 91.3 MiB of resident eight-layer detail K/V, and 4.0 MiB transferred per request across the eight consuming layers. SmolLM2 averages 48.8 KiB, 18.2 MiB, and 0.75 MiB, respectively. Gemma averages a 46.4 KiB index, 5.79 MiB of two-layer detail K/V, and 0.246 MiB transferred; mean TTFT/TPOT are 0.452/0.154 seconds. Mean wall time for

16 routed output tokens is 2.15 seconds for Qwen and 1.85 seconds for SmolLM2, with no native-limit violations in any family. These values describe an eager Python implementation with exact Torch search. They are not evidence of serving speedup: there is no matched dense long-context baseline, fused kernel, asynchronous transfer, or concurrency study.

The public implementation makes the tested boundary reusable. It preserves native GQA storage rather than permanently expanding K/V heads, reports both requested and actual materialization, and permits exact disabled-path testing for a new family. The release includes examples, router save/load artifacts, the model-family notebook, package and CLI tests, and a passing repository test suite. The thin Llama adapter resolves installed rotary and eager-attention functions while the shared core retains publication, routing, budgeting, and replay. Full code-level guidance appears in Appendix J.

D Extended Limitations and Experimental Receipts

The experiments establish mechanisms on modest cohorts, not benchmark accuracy. QASPER provides positive one-shot evidence in sparse discovery, routed likelihood, decoded decision quality, and blind behavioral preference; WikiText-2 verifies ordinary-path preservation; HotpotQA exposes the multi-hop boundary. The primary representation study uses 16 matched examples, the learned router has 32 held-out test identities, and the oracle depth study uses four examples per dataset. The final adaptation study improves identity hygiene—12 HotpotQA train, four validation identities per dataset, and eight test identities per dataset—but is still small. Five seeds measure optimization variation on fixed test identities; they are not five independent dataset samples. The 1,024-token WikiText-2 control can detect drift but is not a broad language-capability evaluation.

The behavioral cohort is also small: 16 identities underlie 152 comparisons because adaptation is repeated over five seeds and each pair is shown twice for reversal. The two judges agree strongly on QASPER but not HotpotQA; one has imperfect identical-answer calibration. Behavioral judging is therefore an additional instrument, not ground truth. The package contains one routed operating point, so it cannot establish where behavioral utility saturates as active K/V changes.

Evidence labels and answer behavior measure different things. Any-evidence recall can be high while a multi-hop example still lacks another required passage. Evidence-identity recall is stricter, but it still assumes that annotations identify the state this checkpoint needs. The positive HotpotQA direct-text control makes its causal comparison interpretable. QASPER's routed likelihood gains remain meaningful paired effects, but direct text lowers target likelihood on the selected causal subset and full sources exceed the matched 2,048-token control budget. QASPER now also shows positive routed decoded evidence under residual 16: polarity, containment, completion, likelihood, and blind behavioral preference all improve over frozen routed PRA. The eight fixed test identities are nevertheless too few to establish benchmark-level QA accuracy or a direct/full-context recovery ratio.

The router is domain-dependent and memory exposure is depth-sensitive. On QASPER, replacing routed memory with oracle memory moves the frozen consumer from 25.0% to 62.5% polarity, while residual 16 reaches $72.5 \pm 5.6\%$. This separates a selection bottleneck from a memory-consumer bottleneck without assigning their small-cohort differences independent additive effects. The routed-trained sequence control learns fluent, terminating answers but reaches only 55.0% polarity, and the validation-selected decision-aware loss falls to 40.0% on untouched test. This small-data control provides no evidence that simple first-token reweighting improves out-of-sample polarity; it does not establish a general limitation of decision-aware objectives or conditional adaptation. Cross-domain any-evidence recall@3 is 0.213 in both directions. Additional gist resolution, centered-RoPE features, query pooling, width, and negative-mining variants do not remove this limitation. All-layer native-K/V injection is also harmful despite using more memory. A larger study should separate calibration, model selection, and test identities. The final study does so, but its oracle-trained adapters remain brittle to routed selections: more M_ψ capacity reduces routed HotpotQA recovery. Future adaptation should train against realistic routing errors and report any-, identity-, and all-evidence recall against both selected-parent and exact-K/V fractions.

The systems result is bounded execution, not serving speed. Detail K/V remains CPU-resident, search is eager and exact, and selected rows are padded. Fused attention, approximate indexing, asynchronous transfer, batching, concurrency, and a matched dense long-context baseline remain unmeasured. CPU residency is not free: reference encoding, host storage, selection, and CPU-to-GPU transfer remain in the measured path. SmolLM2-135M is reported only as a measured Llama-family control; the official Meta Llama 3.2-1B checkpoint remains unmeasured. Gemma 3 uses the official pinned checkpoint, but its five-seed test has only 16 identities per dataset, its systems demonstration has four examples, and its causal probe has eight. Those results validate the integration boundary and causal activation, not benchmark QA quality or calibrated memory use.

The tested checkpoints establish portability through approximately one-billion-parameter scale across multiple decoder families. They do not establish 7B–70B scaling, production throughput, or the economics of a serving-optimized implementation; those require larger hardware, matched dense baselines, and fused transfer/attention measurements.

Paper 2 stops at a successful measured one-shot QASPER boundary. HotpotQA outputs are often semantically near the relevant relation but fail to reach the endpoint entity, consistent with a one-shot retrieval architecture that lacks explicit associative closure. Combined with incomplete discovery, routed adapter brittleness, a likelihood–decoding disconnect, and judge disagreement, this motivates testing whether the boundary lies in entry discovery, associative propagation, or bounded competition rather than further one-shot top-k or oracle-adaptation tuning. Paper 2.5 owns that test while keeping the Paper-2 memory-consumption path fixed; serving optimization and partial materialization remain separate questions.

E Detailed Related Work and Series Ownership

This section locates Paper 2 by the boundary each neighboring architecture changes. The richer literature map appears in the companion survey; the purpose here is self-containment around pretrained retrofit, sparse native-K/V use, and the eight-stage evaluation in Table 3. Availability, retrieval recall, representation, materialization, causal likelihood, task decisions, lexical output, and behavioral preference are not interchangeable endpoints.

E.1 Pretrained retrofit and parameter-efficient adaptation

Full fine-tuning changes the host model, conventional adapters insert trainable modules while freezing the backbone, and LoRA adds low-rank updates to selected pretrained weights [16, 17]. PRA-HF has a different primary intervention: it adds an inference-time memory source to existing attention. Its conservative path does not require updating host weights. A small router learns which external identities to activate; optional residual or LoRA capacity is conditional on memory being present. This structure makes exact disabled-path preservation stronger than a regularization objective: with no selected memory, the wrapper delegates to the original module and the memory-use parameters are absent from the computation.

Retrofitting attention is more than attaching a projection. The adapter must preserve each family's native Q/K/V projections, RoPE convention, grouped- or multi-query head layout, additive mask, cache mutation, output projection, dtype, and return contract. Qwen, SmolLM2, and Gemma therefore share publication, routing, budgeting, and materialization code while thin family adapters retain host-native execution. Paper 2 contributes this exact retrofit boundary; it does not claim that PRA-conditional adapters generally outperform standard PEFT.

E.2 Text retrieval versus latent and native-K/V memory

RAG retrieves text and conditions generation on an encoding of that text [23]. REALM and Atlas train retrieval and reader interfaces, while Self-RAG learns when to retrieve and how to critique evidence [14, 19, 1]. Text materialization has a major advantage: pretrained models learned to consume text in context, and the selected evidence remains directly inspectable. Its costs include prompt space and repeated encoding unless serving caches avoid it.

PRA-HF retrieves compact semantic gists but does not ask attention to consume those gists. It restores selected full-detail post-RoPE K and native V produced by the same checkpoint. This can avoid re-encoding warm references and places memory directly inside native attention, but it inherits a harder integration burden: a pretrained decoder was not trained to receive dynamically restored external state under arbitrary sparse selections. The oracle, depth, and adapter results measure that burden. RAG and PRA can also compose: a corpus retriever may discover a document, after which PRA publishes and repeatedly routes its cached chunks.

E.3 Long-context and sparse-attention connectivity

Longformer and BigBird reduce interactions within an already materialized sequence through local, global, or random sparse edges [3, 40]; Routing Transformer makes sparse connectivity content dependent [30]. Position interpolation and related methods instead extend the coordinate range accepted by dense attention [7, 11]. These methods change the visible sequence or its internal graph. They do not by themselves separate a large external logical store from the K/V active in one operation.

PRA places that boundary before attention: sources are published outside the direct sequence, semantic routing selects identities, and only admitted native K/V participates. This introduces a retrieval miss that dense attention avoids. In return, active K/V can remain bounded while logical availability grows. A production comparison must match answer quality and end-to-end latency, including source encoding and misses; Paper 2 establishes bounded execution but does not claim superiority over optimized long-context kernels.

E.4 KV retention, compression, offload, and reuse

PagedAttention virtualizes physical KV allocation for serving [22]. H_2O, SnapKV, and StreamingLLM retain selected positions from an active chronological cache using attention importance, a prompt observation window, or sink-plus-recency structure [41, 24, 36]. KIVI changes bytes per retained state through low-bit quantization [26]; Quest keeps a backing cache and selects pages for a query [33]. Prompt Cache and CacheBlend reuse modular prompt state and repair composition when necessary [12, 38].

Those methods govern allocation, precision, bandwidth, eviction, or repeated prefill. PRA-HF governs semantic admission of named external objects while preserving full-detail selected payload. CPU-resident PRA detail is therefore not a throughput result: it reduces accelerator residency only if transfer and attention are correspondingly sparse, and it still pays host storage, lookup, and transfer. The mechanisms are complementary. Native-K/V objects could be paged or quantized, and selected chunks could use query-aware physical loading. Partial or smart materialization is intentionally deferred to Papers 3/3.5.

E.5 External, recurrent, and test-time neural memory

Transformer-XL and Compressive Transformer carry chronological hidden state across segments; Feedback Attention Memory, Infini-attention, and Titans retain bounded recurrent, associative, or test-time-updated state [8, 29, 18, 28, 2]. Memorizing Transformer and RETRO retrieve neighboring representations or corpus chunks during model computation [35, 6]. Landmark Attention indexes ingested blocks with learned landmarks [27].

These systems differ in whether memory is chronological or random access, anonymous or named, lossy or full detail, learned jointly or added after pretraining, and consumed through recurrent state, cross-attention, or native attention. PRA-HF retains explicit object identity and layer-native payload and can remove or version an object independently. That inspectability costs storage and requires an external lifecycle. Its selected state participates in the host's native attention operation; its semantic index does not.

E.6 Semantic routers and positional geometry

Content-based sparse attention, dense passage retrieval, Landmark Attention, ClusterKV, and Unlimiformer all use compact or indexed representations to narrow computation [30, 20, 27, 25, 5]. The central Paper 2 distinction is that a representation useful for semantic discovery need not be the representation native attention should consume.

Paper 1.5 establishes that source coordinates, contextual fragmentation, native Q/K ranking, and semantic relevance are separate geometries [31]. Paper 2 transfers that result to pretrained checkpoints: the learned projection ranks compact hidden-state gists, while fixed selected identities recover exact layer-native post-RoPE K and native V. The learned router is therefore pretrained-transfer evidence for a general search/execution separation, not an isolated embedding trick or a replacement attention representation.

E.7 Adaptive computation and depth-specific use

Adaptive Computation Time and Universal Transformer vary the number of representation updates assigned to an input or position [13, 10]. Layerwise cache policies and early-exit systems likewise exploit nonuniform depth. PRA-HF varies a different resource: which external memory is exposed and at which pretrained layers it enters. The measured late-band optimum is consistent with layer specialization, but it does not imply a general lower-versus-upper-layer law. All-layer replay being harmful shows that memory exposure is not monotonically useful. It does not establish a cognitive or biological hierarchy.

E.8 Multi-hop retrieval and the HotpotQA boundary

HotpotQA was designed around multiple supporting facts [37]. One-shot retrieve-then-read can fail when an initial passage reveals the entity or relation needed to form the next query. IRCoT addresses this dependency by alternating reasoning and retrieval; ReAct interleaves reasoning with external actions [34, 39]. These systems pay serial model/retrieval cost but can expand an evidence set in response to intermediate content.

Paper 2 does not implement that loop. Its one-shot router selects from the initial query, then reuses selected identities across a fixed layer band. The relation-near HotpotQA failures and the gap between any-evidence and all-evidence recall are therefore interpreted as a boundary, not as evidence for iterative PRA. Paper 2.5 owns the test of bounded associative closure $q \to A \to B$, with a fixed final unique-memory budget and preserved evidence paths. QASPER remains the one-shot positive control.

E.9 Likelihood, decoding, and behavioral utility

Teacher forcing optimizes next-token likelihood under gold prefixes, whereas generation conditions on model-produced prefixes. Scheduled sampling was proposed to reduce this mismatch [4]; later work also finds meaningful self-recovery and cautions against treating exposure bias as a universal explanation [15]. Paper 2's distinction between answer-critical polarity, completion, lexical match, and blind behavioral preference belongs to this broader training–generation problem, but the paper is not a study of sequence objectives. The failed validation-selected decision-token reweighting is a falsification control for one small-data intervention, not evidence against decision-aware or sequence-level training in general.

Table 26: Related mechanisms alter different boundaries. Entries describe canonical designs; individual systems may compose several rows.

Family	Search or control unit	Consumed representation	Primary distinction from PRA-HF
Dense/sparse context	token position or edge	local native K/V	Evidence is already in the sequence; no external publication boundary
Text RAG	corpus passage	text or encoder state	Familiar text interface; selected evidence is re-encoded or cached
KV management	token/page/cache module	retained or reused K/V	Controls residency, precision, or reuse after encoding, rather than semantic object discovery
Recurrent/neural memory	chronological or learned key	compressed latent/state	Bounded anonymous state rather than independently versioned full-detail objects
Latent/block retrieval	landmark, hidden block, or neighbor	hidden/native state	Closest search analogue; object lifecycle and exact host retrofit differ
Iterative retrieval/agents	generated query or action	text/observation per hop	Discovers follow-up evidence serially; PRA-HF is one-shot
PRA-HF	semantic URI/parent gist	selected exact native K/V	Exact disabled path, named backing objects, and a hard active-K/V budget

E.10 Evaluation stages and cross-paper ownership

Equation 6 provides the canonical eight-stage hierarchy. Reference lifecycle and disabled-path checks test *availability*; retriever and router recall test *discovery*; aligned K/V comparisons test *representation*; cache, span, and attention-support audits test *materialization*; oracle likelihood and depth sweeps test *integration*; polarity and margin test *decision*; completion, containment, EOS, F1, and EM test *decoding*; and blind pairwise judgments test *behavioral utility*. Conditional residual/LoRA training is an intervention on these stages, not an additional stage. A system may report one stage without establishing the next.

The paper series divides ownership accordingly. Paper 1 establishes bounded native-K/V mechanics and logical-versus-active memory [32]. Paper 1.5 owns positional continuity and attention-versus-semantic geometry [31]. This Paper 2 owns exact pretrained retrofit, sparse discovery transfer, depth-dependent causal use, adaptation, decision, decoding, and behavioral diagnostics, and cross-family SDK portability. Paper 2.5 owns iterative associative closure. Papers 3/3.5 own partial and smart materialization. These are interfaces between papers, not claims that the later mechanisms have already been validated.

For a user, PRA-HF provides a Python interface to publish named sources once, route compact semantic state, and activate selected host-native K/V while retaining an exact no-memory path. Ordinary RAG provides a more familiar text interface; dense or sparse context keeps evidence in the sequence; KV compression and offload reduce physical storage or transfer; iterative agents can discover follow-up sources. The SDK is usable at the tested boundary. General production superiority, optimized serving economics, broader task coverage, and larger-model scaling are not established.

F Extended Scientific Synthesis

PRA-HF establishes a cross-family, exact-disabled-path mechanism for externalizing pretrained native K/V, semantically routing compact memory indexes, and reintroducing selected full-detail state under bounded execution. The learned Qwen router moves the recall–sparsity frontier, requested parent fraction contracts to roughly 6–7% physical K/V in the measured product path, and causal use occupies a depth-dependent compatibility band. Thin Qwen, Llama-family, and Gemma adapters preserve their measured host semantics; Gemma additionally validates architecture-informed placement through its native global layers.

The three dataset regimes clarify what this mechanism presently achieves. WikiText-2 shows that PRA-conditional specialization preserves ordinary no-memory behavior. QASPER is the one-shot success regime: decoded polarity progresses from 25.0% routed frozen to 62.5% oracle frozen and $72.5 \pm 5.6\%$ with residual 16, while blind judges independently favor residual adaptation. Selection quality therefore changes decision quality, and memory-use adaptation contributes a further measured gain, without treating those differences as additive causal effects. The routed sequence control shows the remaining distinction: answer realization can improve strongly while decision generalization remains weak. Its failed validation-selected token-reweighting control narrows that explanation but is not the QASPER headline.

HotpotQA marks the one-shot relational boundary: outputs commonly approach a relevant relation but fail to complete it to the endpoint. Clean memory raises the integration ceiling, yet routed adaptation is brittle and behavioral direction is evaluator-dependent. These findings motivate bounded iterative associative closure in Paper 2.5. The cross-family retrofit, bounded native-K/V execution, and public SDK remain Paper 2's engineering result; frozen PRA plus the learned router is the conservative global default, with residual 16 retained as the strongest QASPER-specific candidate and rank-32 LoRA as research-only.

The released `pra_hf` wheel, family adapters, notebook, router artifacts, metrics, frozen judge package and responses, scorer, and reimplementation tests make the claims independently testable. The result is a usable one-shot pretrained-memory boundary with its successes and next architectural step stated separately.

G Routing Ablations

This appendix records the secondary interventions that shaped the final design. They are useful negative results: each changes one plausible source of routing error while leaving native detail K/V unchanged.

The matched 16-example centered-RoPE control makes the dissociation explicit. For $G = 1/2/4/8$, native token-Q/K maximum-rank correlation is .521/.602/.709/.874, while any-evidence R@3 is .188/.125/.250/.250 and routing-index overhead is 1.57/3.14/6.28/12.56% of detail K/V.

Table 27: Parameter-free diagnostics with 32-token parents. Cohorts differ as described in the artifact metadata, so rows summarize directions rather than a single paired leaderboard.

Intervention	Sparse result	Cost or control	Interpretation
Hidden segment means, $G = 1/2/4/8$	Any R@3 0.391/0.313/0.266/0.281	Index overhead 1.57/3.14/6.28/12.56%	Finer untrained means do not improve sparse ranking
Token-level hidden maximum	Any R@3 0.375 versus mean 0.438	Index overhead 50% versus 1.57%	Resolution helps some Hotpot ranks but hurts QASPER
Centered-RoPE, $G = 8$	Any R@3 0.250	Token-QK order correlation 0.874	Coordinate repair does not create evidence semantics
Question decay, strongest tested	Any R@3/MRR 0.250/0.279	Last state 0.469/0.413	More query tokens are not automatically a better query
Width 256 and harder negatives	No validation-selected gain	Twice the 128-wide index at width 256	Data and objective generalization dominate raw capacity

More faithful token-Q/K ordering therefore does not produce monotone semantic-recall improvement. A separate fractional-center control gives R@3 .125/.125/.125 and R@16 .375/.250/.375 for exact/floor/ceil center policies. The rounding convention changes neither the direction nor the interpretation of the result. Full per-example and aggregate records are retained in `docs/papers/shared/results/paper2_hf/routing/centered_rope/`.

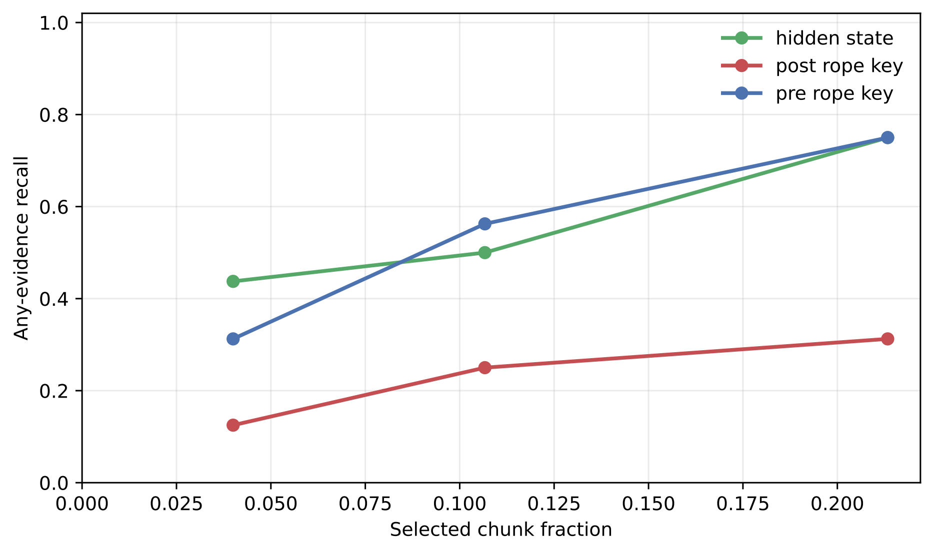

Figure 13: Qwen parameter-free routing diagnostics. Semantic hidden-state and pre-RoPE summaries dominate the post-RoPE key mean for evidence selection, while every fixed selected identity still materializes the same native post-RoPE K/V. This 16-example transfer control is smaller than the learned-router cohort and is not a paired leaderboard with it.

The chunk-size sweep supplies a related caution. At 128-token chunks and $k = 8$, recall reaches 0.688 but the selected fraction is 41.3%; at $k = 16$, 71.5% of parents are selected. Coarser chunks can improve apparent fixed-k recall by making each choice cover more source text. That is not a sparse win, which is why the main paper reports selected fractions and physical K/V.

Independent hidden-state confirmation over 64 evaluations gives any/all-evidence recall of 0.391/0.063 at $k = 3$, 0.578/0.203 at $k = 8$, and 0.797/0.500 at $k = 16$. The selected-parent fractions are 0.045, 0.119, and 0.238, while the hard materialization budget holds the latter two near 0.059 physically active K/V. This illustrates all three distinctions at once: any versus all evidence, selection versus materialization, and fixed rank versus selected fraction.

H QASPER Completion and Decision-Loss Receipts

The 32-token diagnostic is separate from the frozen eight-token behavioral package. It changes no model, prompt, memory, router, or checkpoint and does not rescore the blind comparisons. Its role is to remove incomplete-output confounding and expose decision-specific behavior.

Table 28: Eight-token completion proxy versus the separate 32-token replay. "Old punct." is terminal punctuation in the frozen short output; EOS and cap refer to the diagnostic replay. Adapter rows aggregate optimization replicates over the same eight identities.

Dataset	Condition	Old punct.	New EOS	New cap	Contain.
QASPER	frozen routed	.125	.875	.125	.500
QASPER	residual 16	.200	.950	.050	.825
HotpotQA	frozen routed	.125	.750	.250	.250
HotpotQA	residual 16	.100	.900	.100	.125

For the bounded loss control, standard teacher-forced training uses

$$\mathcal{L}_{\text{seq}} = -\frac{1}{T}\sum_{t=1}^{T} \log p(y_t \mid y_{<t}^{\text{gold}}, x, M), \tag{16}$$

and the intervention adds grouped binary cross-entropy over four tokenizer-native one-token forms for each of yes and no:

$$\mathcal{L} = \mathcal{L}_{\text{seq}} + \lambda \mathcal{L}_{\text{polarity}}. \tag{17}$$

The model, router, routed references, last-14 residual width, optimizer, learning rate, 32 updates, generation length, and seeds remain fixed. Train, validation, and untouched test contain 12, four, and eight independent identities, with yes/no counts 7/5, 1/3, and 5/3. Their majority baselines are therefore 58.3%, 75.0%, and 62.5%. The five seeds are optimization replicates over these fixed identities, not additional documents.

Table 29: Validation-only loss selection. Each value is mean polarity over the four fixed validation identities, followed by standard deviation across five optimization seeds.

λ	0	.25	.5	1	2
Polarity	$.250 \pm .177$	$.200 \pm .209$	$.300 \pm .209$	$.200 \pm .209$	$\mathbf{.400 \pm .137}$

Validation selects $\lambda = 2$ before test is opened. Table 30 retains the seed-level test receipt; both conditions use the same eight identities.

Table 30: Untouched-test polarity by optimization seed. The selected loss does not preserve the sequence-only result.

Condition	11	23	37	53	71	Mean ± seed SD
Sequence only	.500	.375	.625	.625	.625	.550 ± .112
$\lambda = 2$	.250	.375	.500	.250	.625	.400 ± .163

The $\lambda = 0$ run exactly reproduces the prior sequence-only baseline for all 40 seed–item rows: generated text and 12 reported generation, decision, and probability fields have zero mismatches. For the selected adapter, test margin accuracy is 37.5% with no memory, 40.0% with routed memory, and 45.0% with oracle memory; this aggregate ordering is suggestive but every condition remains below the test majority baseline. All 40 no-memory checks are exact. Two 128-token WikiText-2 blocks give loss 3.454496 for the frozen and each adapted no-memory path, a delta of 0.0000 for every seed. Canonical JSON, full per-item rows, aggregate CSVs, and the detailed validation/test curve in PDF and PNG formats are retained under `docs/papers/shared/results/paper2_hf/decision_aware_loss/`.

I Behavioral-Judge Protocol and Audit Record

The behavioral package was frozen before external responses were unblinded. It was generated at repository commit `f86ebe1` with randomization seed 1234 and contains 152 underlying comparisons, each shown in exact A/B-reversed order, for 304 blind items. The displayed item has only an opaque ID, task type, prompt, answers A/B, and requested score names. The private mapping retains condition labels, dataset, source identity, model revision, generation seed, realized routing/materialization fractions, and prompt/answer hashes. Neither package nor mapping was regenerated after judging began.

Table 31: Frozen behavioral-evaluation files. SHA-256 values hash the file bytes. Raw judge responses remain separate from the derived aggregate.

Artifact	SHA-256
Prompt	`65f810c43576ce9ac8eab640c0f8e0ec0acde3d6cfd73c98ccb065eeaf67b9e1`
Single-file blind package	`f66b6030f5a490a2386c1a596efcdb51ea2de7213b51b35c182be92460f65f84`
Private truth mapping	`fe6f3b762ff74053905274e89dc3019e4fc002ec660919609815441197b51453`
GPT-5.6 Sol response	`1260ef7e8a10a61b9730f98d7047fbf577ec3d5e0d2acc6b7576fd6e04ad3ae7`
Claude Sonnet 5 response	`200626052edb6e628fc2cfd47397b834a493859c0386b0448f192097485c3748`
Derived aggregate	`0d70ac5070a023273eaaeecb47852c0f616ceed5366d0722bdf8ac2e2207bb84`

Separate validation receipts and unblinded pair tables are stored under `docs/papers/shared/results/paper2_hf/behavioral_judge/derived/`. Their SHA-256 values are `07a038f3` and `8b82dcb7` for GPT-5.6 Sol validation/pairs, and `1699c6e3` and `ce1ae62e` for Claude Sonnet 5. These are derived outputs; the raw responses above are never overwritten.

The response schema requires one unique record for every opaque ID: semantic equivalence $[0, 100]$, relative quality $[-100, 100]$ (positive favors displayed B), validity A/B $[0, 100]$, confidence $[0, 100]$, and a reason of at most 40 words. Identical-answer and deterministic truncation/contradiction controls test calibration. The scorer first validates IDs, ranges, and completeness; unblinds condition order; flips relative-quality sign when the target was displayed as A; and averages the two orderings into one pair. Thus reversal is a consistency diagnostic, not a second independent observation.

Table 32: Frozen comparison inventory before A/B duplication. All generations are greedy.

Comparison group	Pairs	Purpose
Native no context vs. frozen routed PRA	16	Behavioral change from sparse memory
Native direct evidence vs. frozen routed PRA	16	Gap to concise text evidence
Native full context vs. frozen routed PRA	8	Gap to feasible dense source context
Frozen routed PRA vs. residual-16 PRA	80	Adaptation over 16 identities and five seeds
Identical-answer calibration	16	Positive equivalence anchor
Controlled-corruption calibration	16	Negative calibration diagnostic
Total	152	304 presentations after reversal

Table 33: Judge-specific pair-collapsed results. Eq. is semantic equivalence; ΔQ is relative quality oriented toward the target; ΔV is target minus comparator validity. Cells give GPT-5.6 Sol / Claude Sonnet 5.

Dataset	Target vs. comparator	n	Eq.	ΔQ	ΔV
QASPER	Residual-16 vs. frozen PRA	40	48.4 / 51.9	+4.25 / +16.10	+3.40 / +8.28
HotpotQA	Residual-16 vs. frozen PRA	40	23.8 / 40.0	-6.98 / +43.73	-5.60 / +22.95
QASPER	Frozen PRA vs. no context	8	52.8 / 59.5	-3.13 / -6.50	-2.50 / -6.63
HotpotQA	Frozen PRA vs. no context	8	74.5 / 56.3	-2.00 / +6.00	-1.63 / +2.75
HotpotQA	Frozen PRA vs. direct evidence	8	48.9 / 40.0	-19.25 / -43.63	-15.50 / -23.50
HotpotQA	Frozen PRA vs. full context	8	36.9 / 43.8	-22.63 / -31.38	-18.25 / -20.63

Both judges are exactly invariant to A/B reversal after orientation. On QASPER adaptation, pair-level correlations are .991 for equivalence and .995 for relative quality; across all 152 pairs they are .845 and .375, respectively. GPT-5.6 Sol gives identical copies equivalence 100 on both datasets; Claude Sonnet 5 gives 63.1 on HotpotQA and 88.0 on QASPER. The deterministic negative control can shorten weak generations into more direct answers and is therefore not a clean corruption instrument. These calibration results are why judges remain separate and behavioral evaluation is treated as an additional measurement instrument rather than ground truth.

J Code-Level Adapter Reimplementation Guide

This appendix gives the minimum code structure needed to reproduce the integration without copying the Qwen implementation. Source references use physical line numbers at repository commit `db50846`. This checkpoint makes attention-input hidden-state routing the explicit default and exposes the fixed-selection materialization boundary. The line references are deliberately revision-pinned because upstream Transformers APIs and local diagnostics will move over time. The multi-layer identity mapping, experiments, tests, and complete artifacts are pinned separately at commit `8f6ef41`; all multi-layer line references below refer to that revision.

J.1 Tensor and ownership contract

A new family should add projection, position, cache, mask, kernel, and output conventions in its adapter, while leaving selection and memory policy in the shared core.

The canonical tensor contract is

$$Q \in \mathbb{R}^{B \times H_q \times T_q \times d_h}, \qquad K, V \in \mathbb{R}^{B \times H_{kv} \times T_k \times d_h}. \tag{18}$$

Reference entries use $B = 1$ and keep H_{kv}, including for GQA or MQA. They are not expanded to H_q in persistent storage. The controlled replay implementation shows the only required expansion point in `memory_batching.py`, lines 216–290: after memory and local K/V are joined, native K/V heads may be repeated for the attention computation. Qwen's installed eager kernel performs the corresponding operation in the production adapter.

J.2 Minimal forward-path skeleton

The following pseudocode is the adapter's essential control flow. It is intentionally family-neutral; "native" operations must call the installed model implementation rather than reimplementing its mathematics.

```
01  forward(hidden, native_positions, native_mask, past):
02      if capture_is_armed:
03          q_pre, k_pre, v = native_project_and_norm(hidden)
04          q_post, k_post = native_position(q_pre, k_pre)
05          save_transient_routing_features(hidden, q_pre, q_post, k_pre)
06          save_post_position_detail_kv(k_post, v)
07
08      if memory_is_disabled or reference_cache_is_empty:
09          return original_attention(hidden, native_positions,
10                                    native_mask, past)
11
12      q_pre, k_pre, v = native_project_and_norm(hidden)
13      q_post, k_post = native_position(q_pre, k_pre)
14      if past is not None:
15          k_post, v = past.update(k_post, v, layer_id, native_kwargs)
16      route_q = matched_query(hidden[:, -1, :], q_pre, q_post)
17      memory = shared_core.prepare_memory(q_post, route_q,
18                                          direct_tokens=k_post.shape[2])
19      if memory.is_empty:
20          return native_attention_and_output(q_post, k_post, v, native_mask)
21      k, v, mask = shared_core.combine_local_and_memory_kv(
22          k_post, v, memory, native_mask, query_tokens=q_post.shape[2])
23      return native_attention_and_output(q_post, k, v, mask)
```

Lines 8–10 are a correctness boundary, not an optimization. The Qwen implementation delegates to the untouched module at `hf/qwen.py`, lines 190–201, which makes disabled behavior auditable and bit-exact. In the active path, projection and RoPE occur at lines 204–207 and `Cache.update` occurs once at lines 208–215. Lines 216–221 choose the matched routing query without altering the post-RoPE attention query. If routing returns no materialized memory, lines 224–234 call the native kernel directly. Delegating at that point would execute a second cache update and duplicate generation history.

Memory is composed only after the native cache update. The shared operation at `core.py`, lines 394–436, pads variable memory rows, prepends memory K/V, creates an all-visible additive mask for valid memory positions, masks padding, and concatenates the unchanged local mask. The adapter then invokes Qwen's eager function and one pretrained `o_proj` at `hf/qwen.py`, lines 92–110 and 246–258. Thus local and memory positions share one softmax and one output projection.

J.3 Optional learned routing metric

The learned adapter is a replaceable routing component, not part of native attention. For an asymmetric linear metric,

$$r_q = \mathrm{norm}(W_q h_q), \qquad r_c = \mathrm{norm}(W_c g_c), \qquad s(q,c) = r_q^\top r_c. \tag{19}$$

A shared adapter aliases $W_q = W_c$. An independent implementation needs only the following boundary, revision-pinned at commit `a657bc7`:

```
# Reference publication, after ordinary hidden-state gist pooling
routing_gist = adapter.project_memory(routing_gist)
cache.store(routing_gist, native_post_rope_k, native_v)

# Live routing; native attention continues to use q_post
information_need = aggregate_query_states(h_in, strategy)
routing_query = adapter.project_query(information_need)
selected_ids = cache.search(routing_query)
k_mem, v_mem = cache.materialize_native_kv(selected_ids)
native_attention(q_post, concat(k_mem, k_local),
                 concat(v_mem, v_local))
```

The projection implementation is `hf/routing_adapter.py`, lines 10–74; checkpoint loading and freezing are lines 77–94. Memory projection occurs at `hf/injection.py`, lines 238–246, and live query projection at `hf/qwen.py`, lines 188–202. Cast hidden features to the adapter parameter dtype at this boundary; the checkpoint is FP32 while Qwen emits FP16 states. Do not cast, project, or recompute native detail K/V. Their exact-equality invariant is the test that prevents a routing retrofit from silently becoming an attention rewrite.

J.4 Oracle selection and causal diagnostics

An oracle must replace only identity selection. Adding a separate K/V injection path would confound semantic correctness with different budgeting, positions, masks, or attention code. The operational handle activates a layer set and assigns optional row-local identities in `hf/injection.py`, lines 69–89. The Qwen adapter chooses ordinary routing or the fixed set at `hf/qwen.py`, lines 273–286; both enter `core.py::prepare_selected_memory`, lines 337–388. The latter still applies the trigger, deduplication, whole-chunk budget, overlap policy, residency transfer, and rectangular packing. The fixed-set setter and its ownership contract are in `hf/adapter_base.py`, lines 71–83.

The experiment constructs oracle identities by intersecting annotation token spans with parent logical spans in `run_oracle_memory_use.py`, lines 107–134. Lines 194–281 append the gold continuation, gather its autoregressive token probabilities, retain opt-in eager-attention weights, and use temporary decoder-layer hooks for answer-position hidden states. Attention capture is disabled immediately afterward and is never part of ordinary inference. Conditions and paired deltas are assembled at lines 310–419. A reimplementation should assert three invariants per example: requested oracle IDs equal retained IDs, budget rejection is zero, and the largest native operation stays below the declared bound.

J.5 Route-once multi-layer consumption

Multi-layer PRA shares an identity decision, not an attention tensor. An implementation should make the distinction visible in its types:

```
selected = route_once(query_at_layer_27)       # parent IDs + scores
for layer in consumption_layers:
    layer_hits = []
    for hit in selected:
        chunk = cache[hit.uri].layer[layer].by_id[hit.chunk_id]
        assert chunk.kv.was_projected_by == layer
        layer_hits.append(hit.with_payload(chunk.kv))
    adapter[layer].set_fixed_selection(layer_hits)
model(question)                                # no further semantic search
```

The operational API activates an arbitrary layer set at `hf/injection.py`, lines 70–93. Identity-to-payload resolution is lines 97–145: for each requested layer it looks up the stable `chunk_id` in that layer's `LayerReferenceMemory`, replaces the payload, and records the source routing layer. A missing layer or chunk is an error rather than a fallback to another layer's K/V. The Qwen adapter's fixed-selection branch at `hf/qwen.py`, lines 273–289 then calls the same budgeting, transfer, mask, and attention path as ordinary routing.

The experiment captures the last-layer information-need state and performs one complete ranking at `run_multilayer_pra.py`, lines 161–198. Oracle and routed identities are mapped at lines 277–325; the remainder of the function records per-layer attention, hidden-state changes, materialized tokens, transfers, and latency. The invariant test at `tests/test_hf_integration.py`, lines 548–586 asserts equal parent IDs but different target-layer objects, tensor pointers, and values while preserving native $[B, H_{kv}, T, d_h]$ shape. This catches the most dangerous superficially working implementation: copying $K^{(27)}, V^{(27)}$ into every consuming layer.

J.6 Publishing a reference

Reference construction is a bounded prefill, not a second attention mechanism. A portable implementation follows this recipe:

```
disable_PRA_memory_during_reference_encoding()
for [block_start:block_end] in bounded_encoding_blocks(source):
    position_ids = logical_source_positions(block_start, block_end)
    arm_post_position_kv_capture(position_ids)
    model(block_ids, position_ids=position_ids, use_cache=False)
    captured = consume_native_kv_for_each_PRA_layer()
    for routing_window in overlapping_windows(captured):
        gist = pool(routing_window.attention_input_hidden_state)
        publish(uri, logical_offsets, post_rope_k, native_v, gist)
        discard_token_level_hidden_state()
restore_previous_memory_state()
```

The concrete loop is `hf/injection.py`, lines 235–354. Encoding block size limits one native model invocation; routing chunk size and overlap independently determine index granularity (lines 266–292). Each chunk slices the captured post-RoPE K/V, chooses transient routing features through `_routing_points` (lines 172–233), computes one or more compact gists, and records logical source offsets (lines 293–348). Keeping those two scales separate is important: changing index granularity must not silently change how the pretrained model encoded the source.

The contiguous multi-gist extension keeps that ownership boundary explicit. For a requested count G, split the routing feature rows into balanced source-order ranges, mean each range, score all resulting $[G, D]$ rows, reduce to one parent score, and preserve the winning local index for diagnostics. Top-k must then operate on parent identities before budgeting; otherwise several winning gists could duplicate one parent K/V payload. The reference implementation is revision-pinned at `0205df8`: pooling is in `gists/segment_mean.py`, lines 11–56, publication remains `hf/injection.py`, lines 163–190, and packed scoring plus max reduction remains `memory.py`, lines 487–593.

The same operation implements displaced long-prompt history. At `hf/injection.py`, lines 357–370, the prefix is published under the reserved `#__head` URI and the direct tail retains continued logical position IDs. Every reference-encoding call and every direct call must satisfy the model-safe native bound even when the logical source is longer.

J.7 Porting another decoder family

A practical port begins by locating seven hooks in the installed family: the attention module inside each decoder layer; Q/K/V projections; any Q/K normalization; positional encoding; generation-cache update and its keyword arguments; native mask convention; and the eager attention plus output projection. Implement the six abstract methods declared at `hf/adapter_base.py`, lines 85–106, then add a narrow type-selection branch analogous to `inject_pra` at `hf/injection.py`, lines 379–428. No checkpoint conversion or parameter copy should be needed.

The recommended validation order is:

1. With memory disabled, require exact logits, hidden states, greedy tokens, and generation- cache lengths against an unwrapped copy (test lines 55–72).

2. Capture a short reference and verify post-position state, logical offsets, device residency, and H_{kv} rather than H_q storage (lines 74–93).

3. Verify matched pre/post capture, representation switching, unchanged post-RoPE detail K/V, GQA grouping, serialization, and device parity (lines 109–214).

4. Supply one selected identity through router and oracle traces; require identical native K/V and attention output (lines 241–280).

5. Compare GQA replay with explicit head repetition on a synthetic tensor (lines 216–239).

6. Exercise an over-limit logical prompt and verify bounded native operations plus continued position IDs (lines 282–313).

7. Generate with active memory and assert that direct cache length grows by exactly one per decoded token (lines 315–330).

Four mistakes deserve explicit tests. Applying position encoding again to stored post-position keys changes their basis. Permanently expanding GQA K/V multiplies memory without adding information. Replacing the family mask with a generic causal mask can alter padding or sliding-window behavior. Finally, calling the wrapped module after manually updating its cache appends the same local token twice. Passing transport and parity tests establishes a faithful adapter; it does not establish that the router selects causally useful evidence, which requires the separate quality controls reported in the main paper.

J.8 Llama port and product API

The release implementation is pinned at commit `809694b`. The common abstract family contract remains in `adapter_base.py`, lines 32–132. Qwen's implementation resolves its installed helper functions through an overridable method at `qwen.py`, lines 38–77. The Llama module at `llama.py`, lines 8–48, subclasses that shared projection/capture/control path, replaces the symbol resolver, and invokes Llama's native eager function without the Qwen sliding-window argument. This is inheritance of a family-neutral tensor contract, not reuse of Qwen parameters. Injection selects Qwen or Llama from the concrete attention module at `injection.py`, lines 380–430.

An independent family port should make the same decision only after comparing installed forward signatures and testing every shared assumption. For Llama those assumptions are: Q/K/V linear outputs reshape to $[B, H, T, d_h]$; RoPE receives the same cosine/sine tuple; cache update accepts the layer index, sine, cosine, and cache position; the eager kernel repeats native K/V heads internally; and output returns token-major states before one native output projection. If any family differs, override that specific abstract method rather than adding a family branch to the router, cache, or budgeter. Tests in `test_hf_llama_integration.py`, lines 51–106, cover exact disabled behavior, native-head reference payloads, active memory, and rollover.

The public layer composes rather than replaces those internals. `pra_hf/config.py`, lines 14–127, validates stable product names, resolves negative layer indices, and translates them to the core configuration. `pra_hf/router.py`, lines 14–110, defines the versioned weights/metadata/model-card artifact and legacy-checkpoint conversion. The high-level model in `pra_hf/model.py`, lines 42–358, owns tokenizer/model loading, router compatibility, reference lifecycle, fraction selection, route-once identity replay, generation, chat, and memory economics. In particular, lines 192–258 flatten complete reference-grouped rankings, apply a global fraction or top-k cutoff, reconstruct typed identities, and map those IDs to each consuming layer's native payload. The CLI in `pra_hf/cli.py`, lines 22–145, is a thin caller of the same API; it does not maintain a second execution path. Product tests at `test_pra_hf_api.py`, lines 66–132, pin config serialization and precedence, router save/load and freezing, add/remove/clear, fraction routing, generation, statistics, and the rule that a router cannot change after gists have been indexed.

J.9 Gemma 3 hybrid-attention port

Gemma requires one additional decision before the ordinary family hooks: determine whether a layer is eligible at all. The helper in `hf/gemma3.py`, lines 8–17, reads the checkpoint's `layer_types` schedule and returns only `full_attention` indices. The adapter constructor at lines 34–60 rejects `is_sliding` modules and republishes `is_sliding=False` on the wrapper. This second assignment is necessary because the parent decoder layer inspects the child before selecting its global or local rotary embedding. Omitting it can silently send a global layer through local positional geometry even if the K/V tensors have plausible shapes.

Projection, per-head Q/K RMS normalization, native RoPE, cache update, and output projection are inherited from the already tested family-neutral path. Lines 63–71 then assert the concrete $[B, H_q, T, d_h]$ query and $[B, H_{kv}, T, d_h]$ payload layout. For Gemma 3 1B, $H_q = 4$, $H_{kv} = 1$, and $d_h = 256$. Lines 73–91 call the installed Gemma eager function with the checkpoint's scaling and logit soft cap, explicitly setting no sliding window because this adapter can exist only at a native global layer. K/V head repetition, if needed by the attention calculation, remains inside that native kernel; cached reference payloads stay one-head MQA. The injector's only family-specific addition is the concrete-class selection at `hf/injection.py`, lines 488–499.

The product layer independently enforces the same boundary. `pra_hf/config.py`, lines

65–120, maps the default final layer to Gemma's final global layer, intersects the default late band with the global schedule, and rejects explicit local-layer requests. This duplication is a public validation boundary, not a second implementation of attention. The pinned runner at `experiments/paper2_hf/gemma/run_gemma3_1b.py`, lines 134–167, audits the physical schedule, heads, RoPE bases, normalization, cache, and masks before injection. Lines 170–366 then gate exact disabled tensors, unchanged local module classes, native post-RoPE CPU K/V, continued source positions, causal-prefix invariance, finite enabled output, and bounded `#_head`. Only after those checks does the orchestration at lines 434–543 create features, train five 128-dimensional routers, exercise the public API, and run one causal probe.

The architecture suite in `tests/test_hf_gemma3_integration.py`, lines 93–262, uses Transformers' actual Gemma classes rather than a mock attention module. It pins six independent contracts: exact hybrid-cache parity; untouched local layers; rejection of invalid placement; native global RoPE and Q/K normalization; one-head post-RoPE reference K/V; and bounded product API plus runner behavior. The official-checkpoint runner adds measured pretrained routing, economics, and causal artifacts; the architecture tests remain independent exactness gates.

J.10 Memory calibration path

The calibration code preserves the same ownership boundary. The registered scalar bank in `hf/memory_gate.py`, lines 19–94, supports fixed, shared, and per-layer values. The residual bank in `hf/residual_adapter.py`, lines 10–111, lazily constructs only the requested bottleneck and enables gradients only for that width. Its zero output projection is what makes Equation 12 initially equal frozen PRA rather than a random perturbation. Lines 27–38 accept $[B, T, D]$ hidden and memory-residual tensors, deliberately discard the ordinary hidden state, and return $d_l + W_{up}\,\mathrm{SiLU}(W_{down} d_l)$ in FP32. This delta-only input prevents the adapter from learning an unrelated ordinary-state path.

The Qwen/Llama forward path computes y^{mem} first. Only when a trainable gate or residual adapter is active does `hf/qwen.py`, lines 365–412, invoke local attention a second time, form d_l, apply the gate and adapter in FP32, and cast the result back to the model dtype. The ordinary fixed-PRA path therefore does not pay for the counterfactual attention. Conditional LoRA is computed from the same pre-W_O memory features at lines 365–374 and added after the residual correction at lines 401–412. Registration and public configuration live in `hf/injection.py`, lines 98–140 and 480–495; adapters hold non-owning references so the model registers one parameter bank rather than duplicating it under every attention wrapper. `memory_use_parameters()` at lines 131–140 is the exclusive optimizer boundary for residual, LoRA, or their combination.

The final reproducible protocol is `experiments/paper2_hf/qa/run_last14_combo.py`. Lines 211–280 freeze and optimize only the selected memory-use owner; lines 302–340 build no-context, direct-text, and feasible full context controls; lines 380–488 evaluate oracle and routed memory with physical K/V accounting. Lines 503–690 form per-seed, cohort-recovery, confidence, and paired-combination summaries; lines 827–1050 enforce disjoint train/validation/test identities and validation-only selection. Unit tests in `tests/test_hf_integration.py`, lines 104–152 and 686–824, pin ownership, shape, initialization, exclusive combination gradients, and exact disabled-PRA bypass. `tests/test_hf_last14_combo.py`, lines 13–83, pins layer selection, ratio validity, combination parsing, and cohort-mean recovery.

J.11 Conditional late-band LoRA path

The LoRA implementation is deliberately separate from generic parameter-efficient fine-tuning. `hf/late_band_lora.py`, lines 10–50, defines one rectangular low-rank delta that accepts the pre-output-projection tensor $[B, T, D_a]$ and returns only a correction $[B, T, D]$. Lines 53–149 own a lazy per-layer bank, activate one rank/configuration at a time, and expose only that configuration's trainable factors. This supports Qwen3's 2048 $\rightarrow$ 1024 projection as well as Llama's square projection without family-specific dimensions in the LoRA class.

Registration occurs once in `hf/injection.py`, lines 467–482. Each attention wrapper holds a non-owning reference to that bank, avoiding duplicate parameter registration. Public rank switching and parameter selection are at lines 110–128. In the active Qwen/Llama path, `hf/qwen.py`, lines 348–374, first performs native attention over local plus selected K/V, flattens heads to $[B, T, D_a]$, applies the frozen W_O, and then adds Equation 13 in FP32 at lines 369–374 and 401–412. Reaching those lines already implies that memory exists. The earlier no-memory return and the disabled wrapper therefore never execute LoRA, which is the structural reason the no-PRA control is bit-exact rather than merely regularized toward parity.

The experiment entry point `run_late_band_lora.py`, lines 73–192, configures one exclusive calibration owner and trains it while evaluating a no-PRA control on every update. Lines 194–293 evaluate all four memory conditions; lines 295–353 plot the aggregate; and lines 355–547 run the fixed five-seed cohort and write raw, seed-level, and aggregate artifacts. Tests in `test_hf_integration.py`, lines 122–152 and 739–790, pin validation, rectangular shape handling, zero initialization, parameter ownership, active gradients, and exact Qwen bypass. `test_hf_llama_integration.py`, lines 124–153, exercises the same ownership and bypass contract through the Llama-family wrapper.

J.12 Bounded LoRA search and portable artifact

The follow-up runner makes the selection boundary explicit. Constants and the Stage-A/B grid are in `run_overnight_lora_sweep.py`, lines 52–103; validation scoring and rank-diverse finalist selection are at lines 110–185. Lines 686–740 execute the single-seed screen, lines 742–782 repeat finalists over five seeds and freeze the validation choice, and lines 851–869 release training state before loading the untouched test identities. The full screen is retained below rather than reporting only its winner.

Table 34: Complete single-seed Stage-A/B conditional-LoRA screen, ranked by the equal-weight held-out oracle score. H and Q denote HotpotQA and QASPER. Stage C repeats three rank-diverse finalists over five seeds.

Stage	Rank	Updates	LR	H $\Delta \log p$	Q $\Delta \log p$	Mean
A	32	64	1.0e-03	+18.664	+6.304	+12.484
B	32	128	1.0e-03	+18.222	+5.970	+12.096
A	32	32	1.0e-03	+18.393	+5.014	+11.704
A	16	32	1.0e-03	+17.194	+5.324	+11.259
B	16	128	2.0e-03	+17.480	+4.363	+10.921
B	8	128	2.0e-03	+18.073	+3.698	+10.886
B	16	128	1.0e-03	+16.623	+4.403	+10.513
A	16	64	1.0e-03	+15.457	+5.193	+10.325
B	32	64	2.0e-03	+17.386	+3.118	+10.252
A	8	32	1.0e-03	+15.116	+5.296	+10.206
B	32	128	2.0e-03	+15.487	+4.857	+10.172
A	4	32	1.0e-03	+15.196	+4.952	+10.074
B	16	128	5.0e-04	+14.813	+4.056	+9.435
B	16	64	5.0e-04	+14.494	+4.341	+9.418
B	32	64	5.0e-04	+14.755	+3.976	+9.366
B	8	64	5.0e-04	+13.828	+4.888	+9.358
A	4	64	1.0e-03	+13.667	+5.011	+9.339
B	8	64	2.0e-03	+13.436	+4.634	+9.035
B	8	128	5.0e-04	+13.203	+4.530	+8.866
B	16	64	2.0e-03	+12.863	+4.806	+8.835
B	8	128	1.0e-03	+13.649	+3.684	+8.666
A	8	64	1.0e-03	+12.516	+4.091	+8.303
B	32	128	5.0e-04	+12.120	+3.523	+7.821

The public artifact separates memory-use weights from the router. `PRAMemoryAdapter` in `src/pra_hf/memory_adapter.py`, lines 12–132, validates the conditional-LoRA metadata, reconstructs the per-layer bank, and saves or loads only its state dictionary. The `PRAForCausalLM` constructors accept `memory_adapter` at `src/pra_hf/model.py`, lines 74–132; compatibility checks and late loading are at lines 159–207. Metadata pins the base revision, layer IDs, rank, compatible-router hash, source checkpoint hash, and the measured limitation. Because the adapter is structurally bypassed when PRA has no selected memory, loading it does not weaken disabled-path exactness. It remains an explicit research option rather than an SDK default.

J.13 Global readout control

The LM-head experiment is intentionally outside the product adapter. The class `LMHeadLoRA` in `run_lm_head_control.py`, lines 51–89, holds a non-owning reference to the frozen tied head and registers only rectangular low-rank factors. The zero output factor makes its initial logits exact. Lines 92–124 restore the native head and final normalization before every variant; full-head controls clone the output matrix so optimization cannot mutate the tied input embeddings.

Lines 126–171 build fixed cached WikiText blocks and compute ordinary no-PRA causal loss. Lines 173–300 train oracle-memory examples and the frozen-logit KL control with sequential backward passes, which preserves the summed objective while bounding activation memory. Lines 302–405 evaluate no memory, routed memory, oracle memory, and direct text against both the current and frozen no-memory baselines. Lines 407–668 aggregate the language and QA controls and join them to the earlier gate, residual, and late-LoRA artifacts; lines 670–891 execute the common cohort. Tests in `test_hf_lm_head_control.py`, lines 9–37, pin zero-initialized LoRA equality, factor-only ownership, and exact-but-untied full-head cloning.

J.14 Revision-pinned source map

Table 35 connects the preceding control flow to the implementation.

Table 35: Source map for an independent adapter implementation.

Responsibility	File and physical lines
Abstract family boundary	`src/pra_torch/hf/adapter_base.py`, 16–106
Qwen projections, RoPE, native kernel, and output	`src/pra_torch/hf/qwen.py`, 78–110; 248–432
Llama native-symbol binding and eager invocation	`src/pra_torch/hf/llama.py`, 8–48
Gemma global-layer policy, native MQA, and eager invocation	`src/pra_torch/hf/gemma3.py`, 8–91
Matched routing/detail capture	`src/pra_torch/hf/qwen.py`, 109–244; `src/pra_torch/hf/injection.py`, 80–210
Disabled and active runtime paths	`src/pra_torch/hf/qwen.py`, 248–432
Routing query and GQA reduction	`src/pra_torch/core.py`, 92–121
Budget, deduplication, transfer, and materialization	`src/pra_torch/core.py`, 155–392
Fixed-selection identity/payload boundary	`src/pra_torch/core.py`, 335–392
Route-once layer-native identity mapping	`src/pra_torch/hf/injection.py`, 70–145
K/V and additive-mask composition	`src/pra_torch/core.py`, 394–436
Reference publication and selected-layer injection	`src/pra_torch/hf/injection.py`, 235–428
Depth schedules and causal protocol	`experiments/paper2_hf/qa/run_multilayer_pra.py`, 62–964
Memory gate and residual calibration	`src/pra_torch/hf/memory_gate.py`, 19–94; `src/pra_torch/hf/residual_adapter.py`, 10–111; `src/pra_torch/hf/qwen.py`, 365–412
Conditional output LoRA	`src/pra_torch/hf/late_band_lora.py`, 10–149; `src/pra_torch/hf/qwen.py`, 365–412
Last-14 five-seed convergence protocol	`experiments/paper2_hf/qa/run_last14_combo.py`, 211–1050
Global readout and ordinary-language control	`experiments/paper2_hf/qa/run_lm_head_control.py`, 51–891; `tests/test_hf_lm_head_control.py`, 9–37
Parity, routing, GQA, head, cache, and selection gates	`tests/test_hf_integration.py`, 80–824; `tests/test_hf_last14_combo.py`, 13–83
Stable config, router artifact, and public API	`src/pra_hf/config.py`, 14–127; `src/pra_hf/router.py`, 14–110; `src/pra_hf/model.py`, 42–358
Llama and product-facing gates	`tests/test_hf_llama_integration.py`, 51–153; `tests/test_pra_hf_api.py`, 66–132
Gemma hybrid-attention and pinned-runner gates	`tests/test_hf_gemma3_integration.py`, 93–262; `experiments/paper2_hf/gemma/run_gemma3_1b.py`, 134–543

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